LAQ2

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Note: This answer has 2 submissions:

1. LAQ2.html file
2. LAQ2 report(Answer3.3)

Q1Use two individual and one consensus methods to perform the bulk RNA deconvolution. And produce heatmaps based on the cellular estimates, with samples clustered based on the profile. Briefly describe the patterns you can see. (Hint: pay attention to the input format depending on the methods you use, e.g., logged or unlogged values) (8 marks)

This notebook runs EPIC, MCPcounter for 2 individual methods and Decosus as the consensus method.

To address this assignment, we used the normalized gene expression matrix derived from the DESeq2 workflow performed in the earlier RNA-seq analysis. DESeq2 applies size-factor normalization to correct for differences in library depth across samples and models count data using a negative binomial framework, producing expression values that are comparable between samples. This normalization is conceptually similar to CPM or RPKM in that it adjusts for sequencing depth, but it is statistically more robust because it accounts for count dispersion and mean–variance relationships inherent to RNA-seq data. Therefore, instead of recalculating TPM or RPKM, we used the DESeq2-normalized expression matrix (linear scale for deconvolution methods and log/variance-stabilized form for GSVA), ensuring methodological consistency with the original analysis and appropriate input scaling for each downstream tool.

Q2Perform student t-test on the cellular estimates between tumour and normal groups for these estimates from the three methods above. List the cell types with the significant differences ($p < 0.05$), and produce boxplots for these significant ones.

```
# Load Required Packages

library(DESeq2)
```

```
## Loading required package: S4Vectors
```

```
## Loading required package: stats4
```

```
## Loading required package: BiocGenerics
```

```
## Loading required package: generics
```

```
##
## Attaching package: 'generics'
```

```
## The following objects are masked from 'package:base':
##
##     as.difftime, as.factor, as.ordered, intersect, is.element, setdiff,
##     setequal, union
```

```
##
## Attaching package: 'BiocGenerics'
```

```
## The following objects are masked from 'package:stats':
##
##     IQR, mad, sd, var, xtabs
```

```
## The following objects are masked from 'package:base':
##
##     anyDuplicated, aperm, append, as.data.frame, basename, cbind,
##     colnames, dirname, do.call, duplicated, eval, evalq, Filter, Find,
##     get, grep, grepl, is.unsorted, lapply, Map, mapply, match, mget,
##     order, paste, pmax, pmax.int, pmin, pmin.int, Position, rank,
##     rbind, Reduce, rownames, sapply, saveRDS, table, tapply, unique,
##     unsplit, which.max, which.min
```

```
##
## Attaching package: 'S4Vectors'
```

```
## The following object is masked from 'package:utils':
##
##     findMatches
```

```
## The following objects are masked from 'package:base':
##
##     expand.grid, I, unname
```

```
## Loading required package: IRanges
```

```
## Loading required package: GenomicRanges
```

```
## Loading required package: Seqinfo
```

```
## Loading required package: SummarizedExperiment
```

```
## Loading required package: MatrixGenerics
```

```
## Loading required package: matrixStats
```

```
##
## Attaching package: 'MatrixGenerics'
```

```
## The following objects are masked from 'package:matrixStats':
##
##     colAlls, colAnyNAs, colAnys, colAvgsPerRowSet, colCollapse,
##     colCounts, colCummaxs, colCummins, colCumprods, colCumsums,
##     colDiffs, colIQRDiffs, colIQRs, colLogSumExps, colMadDiffs,
##     colMads, colMaxs, colMeans2, colMedians, colMins, colOrderStats,
##     colProds, colQuantiles, colRanges, colRanks, colSdDiffs, colSds,
##     colSums2, colTabulates, colVarDiffs, colVars, colWeightedMads,
##     colWeightedMeans, colWeightedMedians, colWeightedSds,
##     colWeightedVars, rowAlls, rowAnyNAs, rowAnys, rowAvgsPerColSet,
##     rowCollapse, rowCounts, rowCummaxs, rowCummins, rowCumprods,
##     rowCumsums, rowDiffs, rowIQRDiffs, rowIQRs, rowLogSumExps,
##     rowMadDiffs, rowMads, rowMaxs, rowMeans2, rowMedians, rowMins,
##     rowOrderStats, rowProds, rowQuantiles, rowRanges, rowRanks,
##     rowSdDiffs, rowSds, rowSums2, rowTabulates, rowVarDiffs, rowVars,
##     rowWeightedMads, rowWeightedMeans, rowWeightedMedians,
##     rowWeightedSds, rowWeightedVars
```

```
## Loading required package: Biobase
```

```
## Welcome to Bioconductor
##
##     Vignettes contain introductory material; view with
##     'browseVignettes()'. To cite Bioconductor, see
##     'citation("Biobase")', and for packages 'citation("pkgname")'.
```

```
##
## Attaching package: 'Biobase'
```

```
## The following object is masked from 'package:MatrixGenerics':
##
##     rowMedians
```

```
## The following objects are masked from 'package:matrixStats':
##
##     anyMissing, rowMedians
```

```
library(dplyr)
```

```
##
## Attaching package: 'dplyr'
```

```
## The following object is masked from 'package:Biobase':
##
##     combine
```

```
## The following object is masked from 'package:matrixStats':
##
##     count
```

```
## The following objects are masked from 'package:GenomicRanges':
##
##     intersect, setdiff, union
```

```
## The following object is masked from 'package:Seqinfo':
##
##     intersect
```

```
## The following objects are masked from 'package:IRanges':
##
##     collapse, desc, intersect, setdiff, slice, union
```

```
## The following objects are masked from 'package:S4Vectors':
##
##     first, intersect, rename, setdiff, setequal, union
```

```
## The following objects are masked from 'package:BiocGenerics':
##
##     combine, intersect, setdiff, setequal, union
```

```
## The following object is masked from 'package:generics':
##
##     explain
```

```
## The following objects are masked from 'package:stats':
##
##     filter, lag
```

```
## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```
library(ggplot2)
library(pheatmap)
library(gplots)
```

```
##
## ---------------------
## gplots 3.3.0 loaded:
##   * Use citation('gplots') for citation info.
##   * Homepage: https://talgalili.github.io/gplots/
##   * Report issues: https://github.com/talgalili/gplots/issues
##   * Ask questions: https://stackoverflow.com/questions/tagged/gplots
##   * Suppress this message with: suppressPackageStartupMessages(library(gplots))
## ---------------------
```

```
##
## Attaching package: 'gplots'
```

```
## The following object is masked from 'package:IRanges':
##
##     space
```

```
## The following object is masked from 'package:S4Vectors':
##
##     space
```

```
## The following object is masked from 'package:stats':
##
##     lowess
```

```
# Deconvolution methods
library(EPIC)
library(MCPcounter)
```

```
## Loading required package: curl
```

```
## Using libcurl 8.14.1 with LibreSSL/3.3.6
```

```
library(Decosus)

# GSVA & gene set analysis
library(GSVA)


# Parallel backend (serial for reproducibility)
library(BiocParallel)
register(SerialParam())
```

```
# Load Raw Counts and Sample Data


rawCounts <- read.delim(
  "/Users/dr.parul/Downloads/data_required_for_CAN7032-7132/LAQ2_data/row_count_data.nodup.tx
t",
  header = TRUE,
  row.names = 1,
  check.names = FALSE
)  # Reads the raw RNA-seq count matrix (genes × samples)

sampleData <- read.delim(
  "/Users/dr.parul/Downloads/data_required_for_CAN7032-7132/LAQ2_data/sample_groups.txt",
  header = TRUE,
  row.names = 1
)  # Reads sample metadata (Patient ID and Group: Normal/Tumour)

# Ensure correct sample order
sampleData <- sampleData[colnames(rawCounts), ]
# Reorders metadata rows so they match the column order of the count matrix

# Convert variables to factors
sampleData$Patient <- factor(sampleData$Patient)
# Converts Patient column into a categorical variable

sampleData$Group <- factor(sampleData$Group)
# Converts Group (Tumour/Normal) into a categorical variable

sampleData$Group <- relevel(sampleData$Group, ref = "Normal")
# Sets "Normal" as the reference level for comparisons

# Create DESeq2 Object


dds <- DESeqDataSetFromMatrix(
  countData = rawCounts,
  colData   = sampleData,
  design    = ~ Patient + Group
)
# Creates DESeq2 dataset with paired design (controls for Patient effect)

# Filter low-count genes
keep <- rowSums(counts(dds) >= 10) >= 5
# Keeps genes that have at least 10 reads in at least 5 samples

dds <- dds[keep, ]
# Removes low-expression genes to improve statistical stability

# Run DESeq normalization
dds <- DESeq(dds)
```

```
## estimating size factors
```

```
## estimating dispersions
```

```
## gene-wise dispersion estimates
```

```
## mean-dispersion relationship
```

```
## final dispersion estimates
```

```
## fitting model and testing
```

```
# Estimates size factors, dispersion, and fits the negative binomial model
# This produces normalized counts internally and prepares the object for downstream analysis
```

We need to use 1. linear matrix for deconvolution 2. log/VST matrix for GSVA Saving both below:

```
# Extract Normalized Linear Counts



norm_counts <- counts(dds, normalized = TRUE)
norm_counts_df <- data.frame(Gene = rownames(norm_counts),
                             norm_counts,
                             check.names = FALSE)

# Save linear normalized counts
write.table(norm_counts_df,
            "/Users/dr.parul/Downloads/data_required_for_CAN7032-7132/LAQ2_data/deseq_normali
zed_counts_linear.txt",
            sep = "\t",
            quote = FALSE,
            row.names = FALSE)



# Extract VST Matrix (for GSVA)



vst_obj <- vst(dds, blind = FALSE)
vst_mat <- assay(vst_obj)

vst_mat_df <- data.frame(Gene = rownames(vst_mat),
                         vst_mat,
                         check.names = FALSE)

# Save VST matrix
write.table(vst_mat_df,
            "/Users/dr.parul/Downloads/data_required_for_CAN7032-7132/LAQ2_data/deseq_vst_mat
rix.txt",
            sep = "\t",
            quote = FALSE,
            row.names = FALSE)

cat("Normalized linear matrix and VST matrix saved successfully.\n")
```

```
## Normalized linear matrix and VST matrix saved successfully.
```

```
#checking number of genes in the data

cat("Number of genes in normalized linear matrix:",
    nrow(norm_counts), "\n")
```

```
## Number of genes in normalized linear matrix: 12403
```

```
cat("Number of genes in VST matrix:",
    nrow(vst_mat), "\n")
```

```
## Number of genes in VST matrix: 12403
```

```
cat("Number of samples:",
    ncol(norm_counts), "\n")
```

```
## Number of samples: 10
```

Normalization changes the scale and variance structure Different normalizations do different things: TPM normalizes for library size and gene length. DESeq2 size-factor normalization corrects for library size but keeps counts in a count-based framework. VST/rlog additionally stabilizes variance across expression ranges.

DESeq2 normalization is safer for this assignment: DESeq2 normalization: Models count dispersion, Handles mean–variance dependence, Produces more statistically stable inputs for downstream testing. TPM: Is purely scaling-based, Does not account for dispersion, May distort variance structure for GSVA comparisons.

```
norm_mat <- as.matrix(norm_counts)
epic_out <- EPIC(bulk = norm_mat)

est_EPIC_mRNAProp <- t(epic_out$mRNAProportions)
est_EPIC_cellProp <- t(epic_out$cellFractions)

colSums(est_EPIC_mRNAProp)
```

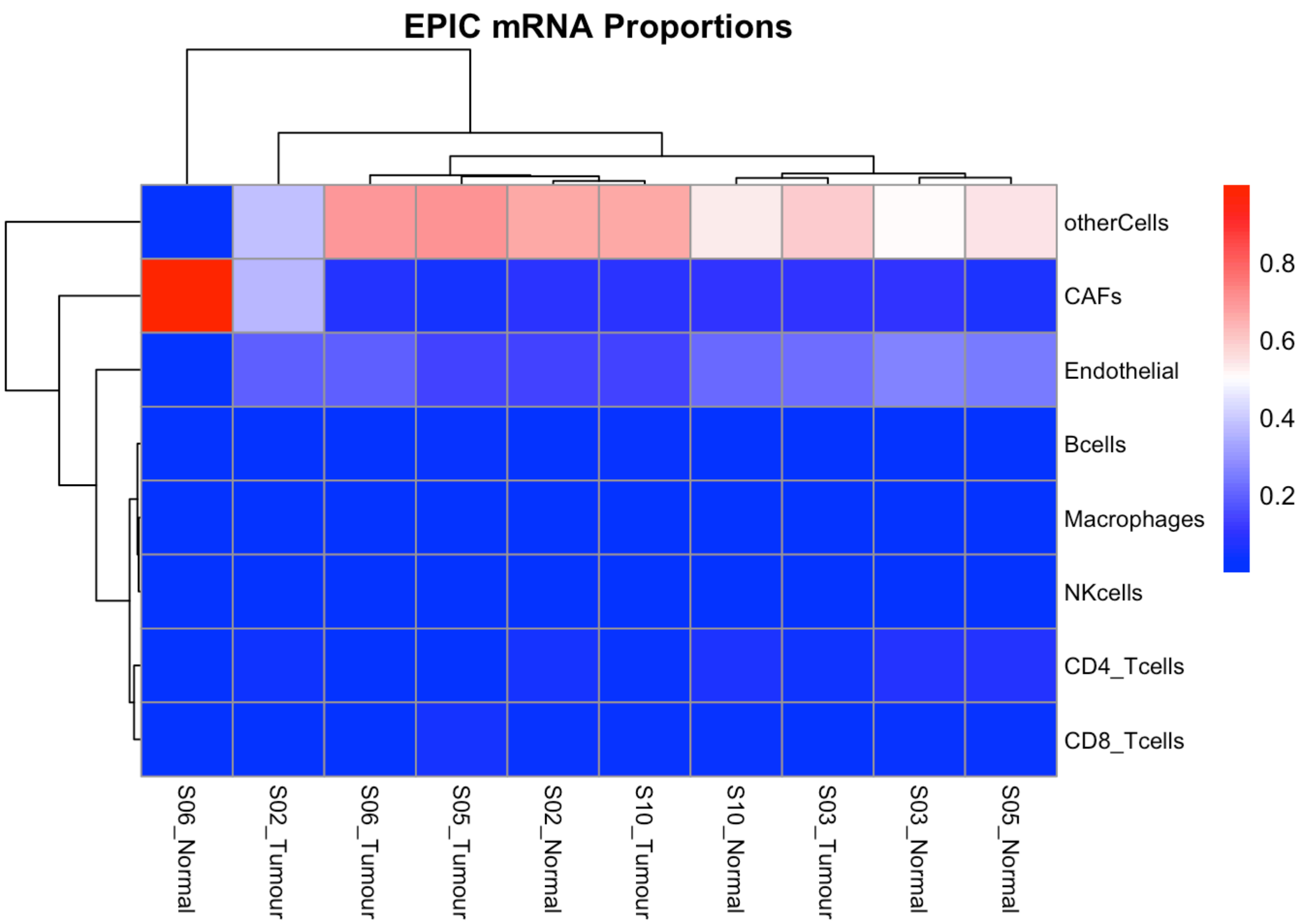
```
## S02_Normal S03_Normal S05_Normal S06_Normal S10_Normal S02_Tumour S03_Tumour
##          1          1          1          1          1          1          1
## S05_Tumour S06_Tumour S10_Tumour
##          1          1          1
```

```
colSums(est_EPIC_cellProp)
```

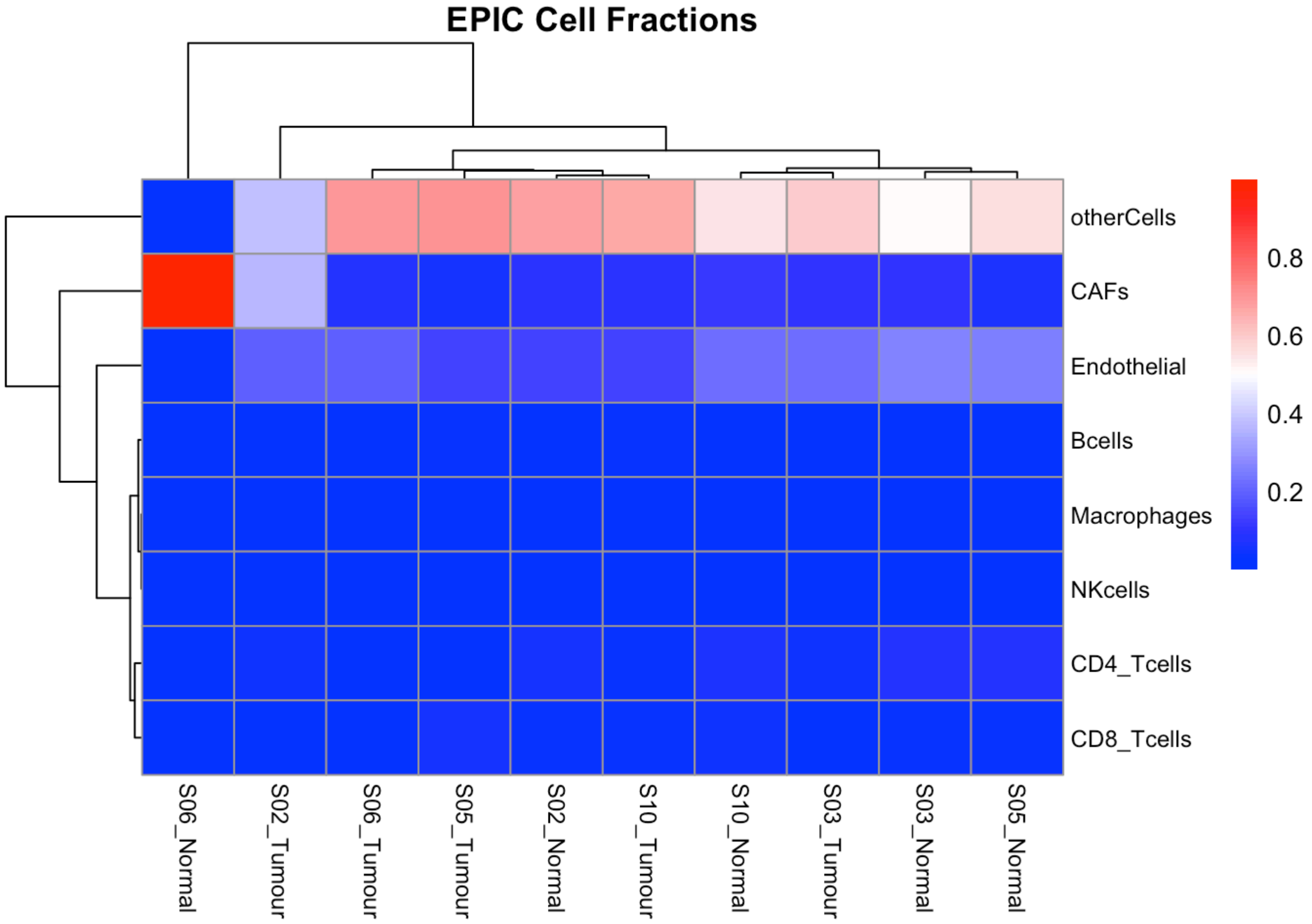
```
## S02_Normal S03_Normal S05_Normal S06_Normal S10_Normal S02_Tumour S03_Tumour
##          1          1          1          1          1          1          1
## S05_Tumour S06_Tumour S10_Tumour
##          1          1          1
```

```
# Helper function to save pheatmap without re-rendering
save_pheatmap <- function(x, filepath) {
  png(filepath, width = 1200, height = 900)
  grid::grid.newpage()
  grid::grid.draw(x$gtable)
  dev.off()
}

# Render in knit AND save separately
p1 <- pheatmap(est_EPIC_mRNAProp,
               color = colorRampPalette(c("blue", "white", "red"))(100),
               scale = "none", cluster_rows = TRUE, cluster_cols = TRUE,
               main = "EPIC mRNA Proportions",
               fontsize_col = 9, fontsize_row = 9)
```




```
p2 <- pheatmap(est_EPIC_cellProp,
               color = colorRampPalette(c("blue", "white", "red"))(100),
               scale = "none", cluster_rows = TRUE, cluster_cols = TRUE,
               main = "EPIC Cell Fractions",
               fontsize_col = 9, fontsize_row = 9)
```




```
save_pheatmap(p1, "/Users/dr.parul/Downloads/data_required_for_CAN7032-7132/LAQ2_data/EPIC_mRNAProportions_2025-02-25.png")
```

```
## quartz_off_screen
##                  2
```

```
save_pheatmap(p2, "/Users/dr.parul/Downloads/data_required_for_CAN7032-7132/LAQ2_data/EPIC_cellFractions_2025-02-25.png")
```

```
## quartz_off_screen
##                  2
```

```
cat("EPIC heatmaps saved successfully.\n")
```

```
## EPIC heatmaps saved successfully.
```

```
est_EPIC_cellProp
```

```
##               S02_Normal   S03_Normal   S05_Normal   S06_Normal   S10_Normal
## Bcells      1.137165e-03 4.381454e-03 1.000067e-02 3.879726e-06 2.059762e-02
## CAFs        9.833110e-02 1.062500e-01 6.153233e-02 9.998532e-01 1.100748e-01
## CD4_Tcells  5.460846e-02 7.465288e-02 8.283862e-02 9.614951e-07 6.266662e-02
## CD8_Tcells  3.615122e-02 3.268413e-02 3.385009e-02 7.186921e-09 4.011587e-02
## Endothelial 1.340629e-01 2.683313e-01 2.523762e-01 1.236296e-05 2.214252e-01
## Macrophages 2.901023e-03 4.187683e-03 6.227283e-03 2.712916e-06 3.081669e-03
## NKcells     3.922007e-09 3.607293e-08 1.240696e-08 1.268551e-04 1.566788e-08
## otherCells  6.728082e-01 5.095126e-01 5.531748e-01 4.403320e-10 5.420382e-01
##               S02_Tumour   S03_Tumour   S05_Tumour   S06_Tumour   S10_Tumour
## Bcells      3.631597e-03 7.830548e-03 0.0307520899 1.031628e-02 3.043507e-02
## CAFs        3.693796e-01 1.066155e-01 0.0538425760 7.919238e-02 9.577289e-02
## CD4_Tcells  4.297281e-02 4.715718e-02 0.0167675813 5.423851e-03 3.752145e-02
## CD8_Tcells  5.152317e-06 1.838821e-02 0.0565899463 1.902714e-02 3.201902e-02
## Endothelial 1.954228e-01 2.268905e-01 0.1346898486 1.912527e-01 1.346966e-01
## Macrophages 4.161355e-03 1.034959e-03 0.0016720017 4.829663e-05 9.868853e-04
## NKcells     7.267669e-07 1.994785e-07 0.0009323928 4.219037e-08 1.901911e-08
## otherCells  3.844259e-01 5.920829e-01 0.7047535633 6.947393e-01 6.685681e-01
```

This is a matrix of: Estimated cell-type proportions per sample Values range from 0 to 1 Columns = samples Rows = cell types Each column sums to ~1 These are the final inferred cellular composition fractions.

EPIC Results Summary:

mRNA Proportions heatmap: otherCells dominates all samples. S06_Normal is a clear outlier with very high CAFs. Immune cells (CD4, CD8, Bcells, NK, Macrophages) are consistently low across all samples. Samples do not cluster cleanly by tumour/normal.

Cell Fractions heatmap: Same pattern — otherCells highest across most samples, S06_Normal extreme outlier for CAFs (red), S02_Tumour also elevated CAFs. Endothelial cells moderately present. Immune infiltration minimal throughout.

Numeric table confirms:

otherCells: 38–70% across all samples (dominant fraction)

CAFs: S06_Normal ~99.9%, all others much lower (~5–36%)

Endothelial: ~13–27% in most samples, near zero in S06_Normal

All immune cell types (Bcells, CD4, CD8, NK, Macrophages): very low, mostly <5%

```
# Running paired Student t-tests (Tumour vs Normal, matched by patient)
epic_df <- as.data.frame(est_EPIC_cellProp)

# Extracting patient ID and group from sample names
sample_names <- colnames(epic_df)
group <- ifelse(grepl("Tumour", sample_names), "Tumour", "Normal")
patient <- gsub("_(Tumour|Normal)", "", sample_names)

# Initialising results storage
t_test_results <- data.frame(
  Cell_Type = rownames(epic_df),
  Mean_Normal = NA,
  Mean_Tumour = NA,
  P_value = NA
)

# Looping through each cell type
for (i in 1:nrow(epic_df)) {

  values <- as.numeric(epic_df[i, ])

  # Building a data frame to ensuring patient-matched ordering
  df_temp <- data.frame(value = values, group = group, patient = patient)

  # Ordering normal and tumour values by patient ID
  normal_ordered <- df_temp[df_temp$group == "Normal", ]
  tumour_ordered <- df_temp[df_temp$group == "Tumour", ]
  normal_ordered <- normal_ordered[order(normal_ordered$patient), ]
  tumour_ordered <- tumour_ordered[order(tumour_ordered$patient), ]

  # Performing paired t-test
  test <- t.test(tumour_ordered$value, normal_ordered$value, paired = TRUE)

  # Storing mean values and p-value
  t_test_results$Mean_Normal[i] <- mean(normal_ordered$value)
  t_test_results$Mean_Tumour[i] <- mean(tumour_ordered$value)
  t_test_results$P_value[i] <- test$p.value
}

# Identifying significant cell types (p < 0.05)
significant_cells <- t_test_results[t_test_results$P_value < 0.05, ]
t_test_results
```

Cell_Type <chr>	Mean_Normal <dbl>	Mean_Tumour <dbl>	P_value <dbl>
Bcells	7.224158e-03	0.016593116	0.04541982
CAFs	2.752083e-01	0.140960593	0.54618723
CD4_Tcells	5.495351e-02	0.029968574	0.10218415
CD8_Tcells	2.856026e-02	0.025205892	0.77469077
Endothelial	1.752416e-01	0.176590505	0.98203594

Cell_Type	Mean_Normal	Mean_Tumour	P_value
<chr>	<dbl>	<dbl>	<dbl>
Macrophages	3.280074e-03	0.001580699	0.18227589
NKcells	2.538463e-05	0.000186676	0.45322513
otherCells	4.555067e-01	0.608913944	0.38426541

8 rows

```
significant_cells
```

	Cell_Type	Mean_Normal	Mean_Tumour	P_value
	<chr>	<dbl>	<dbl>	<dbl>
1	Bcells	0.007224158	0.01659312	0.04541982

1 row

only Bcells shows a significant difference between tumour and normal in EPIC cell fractions.

```
# Ensuring matrix format
norm_mat <- as.matrix(norm_counts)

# Running MCP-counter
est_MCPcounter <- MCPcounter.estimate(
  norm_mat,
  featuresType = "HUGO_symbols"
)

# Checking dimensions
dim(est_MCPcounter)
```

```
## [1]  9 10
```

```
# Splitting into Normal and Tumour tables
mcp_df <- as.data.frame(est_MCPcounter)
mcp_df <- cbind(Cell_Type = rownames(mcp_df), mcp_df)
rownames(mcp_df) <- NULL

# Normal samples only
normal_cols <- c("Cell_Type", colnames(mcp_df)[grepl("Normal", colnames(mcp_df))])
mcp_normal <- mcp_df[, normal_cols]
cat("Normal Samples:\n")
```

```
## Normal Samples:
```

```
mcp_normal
```

Cell_Type <chr>	S02_Normal <dbl>	S03_Nor... <dbl>	S05_Nor... <dbl>	S06_Normal <dbl>	S10_Nor... <dbl>
T cells	52.832942	74.28665	63.84339	57.15523	129.11276
CD8 T cells	41.055735	47.22860	53.73429	133.38723	93.74663
Cytotoxic lymphocytes	32.568252	66.90718	72.66341	35.59997	45.92318
B lineage	3.158133	10.16727	11.66957	17.42446	72.52696
Monocytic lineage	159.260161	179.07510	264.36805	151.41253	162.60879
Myeloid dendritic cells	80.729788	135.78222	66.35371	120.99483	201.74529
Neutrophils	736.860220	250.19912	288.44381	383.03765	327.02735
Endothelial cells	155.044616	204.00131	170.56566	282.54660	189.73664
Fibroblasts	1870.517355	1763.76326	1081.31534	31556.89489	1797.20262

9 rows

```
# Tumour samples only
tumour_cols <- c("Cell_Type", colnames(mcp_df)[grepl("Tumour", colnames(mcp_df))])
mcp_tumour <- mcp_df[, tumour_cols]
cat("Tumour Samples:\n")
```

```
## Tumour Samples:
```

```
mcp_tumour
```

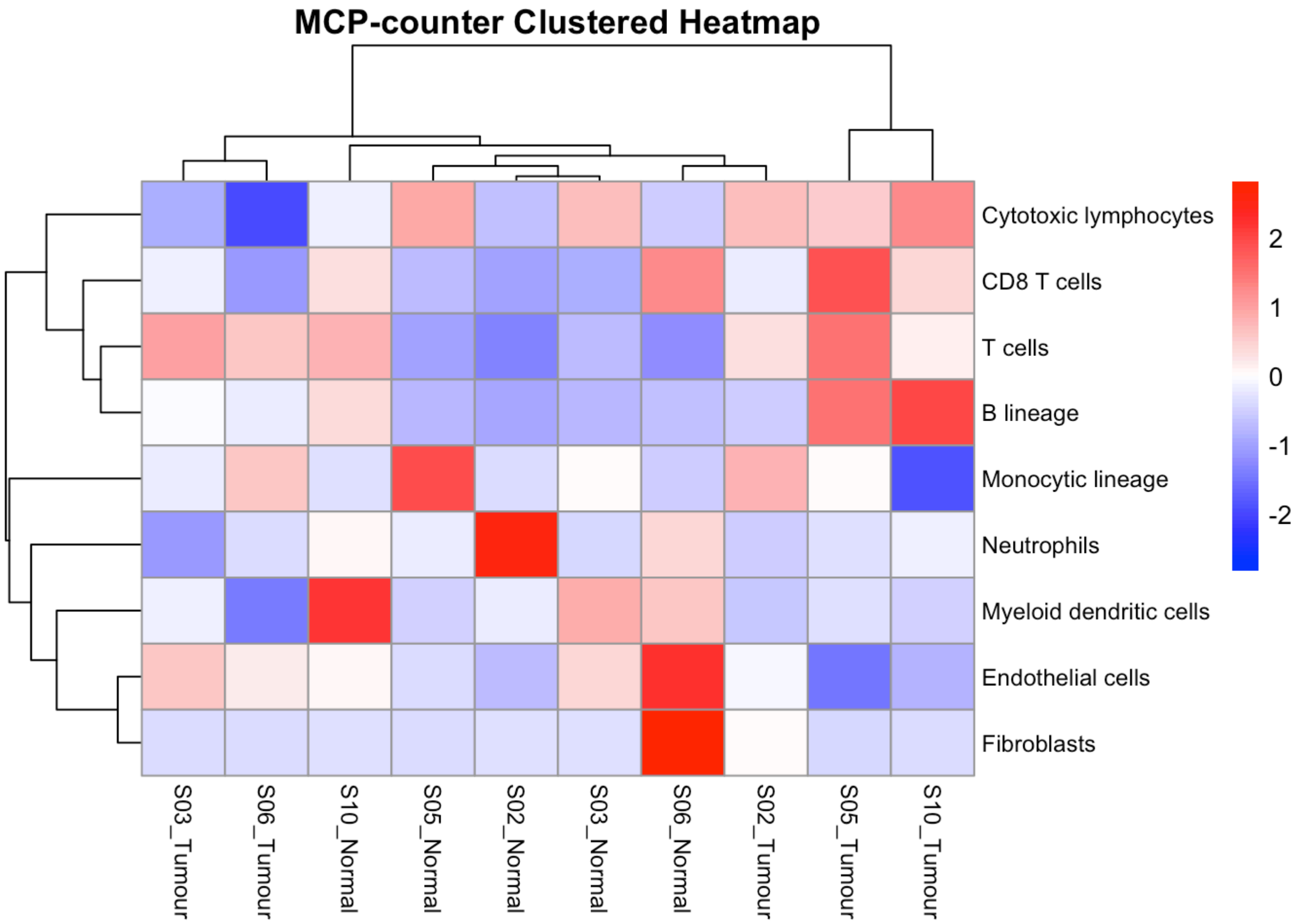
Cell_Type <chr>	S02_Tum... <dbl>	S03_Tum... <dbl>	S05_Tum... <dbl>	S06_Tum... <dbl>	S10_Tum... <dbl>
T cells	109.91096	134.58652	152.04980	119.88926	103.99614
CD8 T cells	74.74877	76.51792	158.69432	37.81996	99.63702
Cytotoxic lymphocytes	67.93755	27.80151	62.70361	0.00000	81.57781
B lineage	24.91626	52.20222	133.01938	43.11475	161.07986
Monocytic lineage	214.96508	166.88194	177.38375	202.93108	89.49540
Myeloid dendritic cells	59.20522	81.36405	73.34774	15.12798	64.45270
Neutrophils	229.03661	135.40028	270.16740	259.98519	290.01491
Endothelial cells	182.50575	211.44451	121.95594	194.39458	151.01236
Fibroblasts	4970.04462	1517.96975	851.64106	1178.46986	1498.11382

9 rows

```
# Clustering columns (samples) by Spearman correlation
hc <- hclust(
  as.dist(1 - cor(est_MCPcounter, method = "spearman")),
  method = "ward.D"
)

# Clustering rows (cell types) by Pearson correlation
hr <- hclust(
  as.dist(1 - cor(t(est_MCPcounter), method = "pearson")),
  method = "average"
)

# Rendering and saving MCP-counter heatmap
p_mcp <- pheatmap(
  as.matrix(est_MCPcounter),
  cluster_rows = hr,
  cluster_cols = hc,
  color = colorRampPalette(c("blue", "white", "red"))(100),
  scale = "row",
  main = "MCP-counter Clustered Heatmap",
  fontsize_col = 9,
  fontsize_row = 9
)
```




```
save_pheatmap(p_mcp, "/Users/dr.parul/Downloads/data_required_for_CAN7032-7132/LAQ2_data/MCPc
ounter_Heatmap_2025-02-25.png")
```

```
## quartz_off_screen
##                 2
```

```
cat("MCP-counter heatmap saved successfully.\n")
```

```
## MCP-counter heatmap saved successfully.
```

```
# Building paired comparison table (Normal vs Tumour per patient)
mcp_mat <- as.data.frame(est_MCPcounter)

# Creating paired table per patient
patients <- c("S02", "S03", "S05", "S06", "S10")

paired_list <- lapply(patients, function(p) {
  normal_col <- paste0(p, "_Normal")
  tumour_col <- paste0(p, "_Tumour")
  data.frame(
    Cell_Type = rownames(mcp_mat),
    Patient = p,
    Normal = mcp_mat[[normal_col]],
    Tumour = mcp_mat[[tumour_col]],
    Difference = mcp_mat[[tumour_col]] - mcp_mat[[normal_col]]
  )
})

paired_mcp <- do.call(rbind, paired_list)
rownames(paired_mcp) <- NULL
paired_mcp
```

Cell_Type <chr>	Patient <chr>	Normal <dbl>	Tumour <dbl>	Difference <dbl>
T cells	S02	52.832942	109.91096	57.078020
CD8 T cells	S02	41.055735	74.74877	33.693033
Cytotoxic lymphocytes	S02	32.568252	67.93755	35.369297
B lineage	S02	3.158133	24.91626	21.758123
Monocytic lineage	S02	159.260161	214.96508	55.704923
Myeloid dendritic cells	S02	80.729788	59.20522	-21.524571
Neutrophils	S02	736.860220	229.03661	-507.823606
Endothelial cells	S02	155.044616	182.50575	27.461138
Fibroblasts	S02	1870.517355	4970.04462	3099.527266
T cells	S03	74.286650	134.58652	60.299866

1-10 of 45 rows Previous **1** 2 3 4 5 Next

MCP-counter Results Summary: Heatmap patterns:

Samples do not cluster cleanly by tumour/normal — mixed grouping S06_Normal is again an outlier, showing extremely high Fibroblasts score (consistent with EPIC's CAFs finding) S03_Tumour and S05_Tumour show elevated T cells, CD8 T cells and Cytotoxic lymphocytes S02_Tumour shows notably high Fibroblasts score Monocytic lineage is broadly elevated across most samples

Table observations:

Fibroblasts dominate — S06_Normal extreme (31556) vs others (~850–4970) Monocytic lineage consistently high across all samples (~60–265) T cells, CD8 T cells, Cytotoxic lymphocytes show more variability between samples B lineage remains low across most samples Myeloid dendritic cells and Neutrophils show moderate scores

Key difference from EPIC: MCP-counter gives absolute enrichment scores (not proportions), so values are not constrained to sum to 1. This explains the large numeric range. The S06_Normal Fibroblasts outlier is consistent across both methods, strengthening confidence in that finding.

```
# Converting MCP matrix to numeric data frame
mcp_df <- as.data.frame(est_MCPcounter)

# Extracting group and patient ID from column names
sample_names <- colnames(mcp_df)
group <- ifelse(grepl("Tumour", sample_names), "Tumour", "Normal")
patient <- gsub("_(Tumour|Normal)", "", sample_names)

# Initialising results storage
mcp_ttest_results <- data.frame(
  Cell_Type = rownames(mcp_df),
  Mean_Normal = NA,
  Mean_Tumour = NA,
  P_value = NA
)

# Looping through each cell type
for (i in 1:nrow(mcp_df)) {

  values <- as.numeric(mcp_df[i, ])

  # Building data frame for patient-matched ordering
  df_temp <- data.frame(value = values, group = group, patient = patient)

  # Ordering normal and tumour values by patient ID
  normal_ordered <- df_temp[df_temp$group == "Normal", ]
  tumour_ordered <- df_temp[df_temp$group == "Tumour", ]
  normal_ordered <- normal_ordered[order(normal_ordered$patient), ]
  tumour_ordered <- tumour_ordered[order(tumour_ordered$patient), ]

  # Performing paired t-test
  test <- t.test(tumour_ordered$value, normal_ordered$value, paired = TRUE)

  # Storing mean values and p-value
  mcp_ttest_results$Mean_Normal[i] <- mean(normal_ordered$value)
  mcp_ttest_results$Mean_Tumour[i] <- mean(tumour_ordered$value)
  mcp_ttest_results$P_value[i] <- test$p.value
}

# Viewing all results
mcp_ttest_results
```

Cell_Type <chr>	Mean_Normal <dbl>	Mean_Tumour <dbl>	P_value <dbl>
T cells	75.44619	124.08654	0.06488821
CD8 T cells	73.83050	89.48360	0.65394196
Cytotoxic lymphocytes	50.73240	48.00410	0.87596280
B lineage	22.98928	82.86649	0.03680390
Monocytic lineage	183.34493	170.33145	0.68658284
Myeloid dendritic cells	121.12117	58.69954	0.07806601

Cell_Type <chr>	Mean_Normal <dbl>	Mean_Tumour <dbl>	P_value <dbl>
Neutrophils	397.11363	236.92088	0.14739223
Endothelial cells	200.37897	172.26263	0.24409347
Fibroblasts	7613.93869	2003.24782	0.41845128

9 rows

```
# Identifying significant cell types (p < 0.05)
mcp_significant <- mcp_ttest_results[mcp_ttest_results$P_value < 0.05, ]
mcp_significant
```

	Cell_Type <chr>	Mean_Normal <dbl>	Mean_Tumour <dbl>	P_value <dbl>
4	B lineage	22.98928	82.86649	0.0368039

1 row

MCP-counter paired t-test results: Only B lineage is significant (p = 0.037) — Mean Normal = 22.99, Mean Tumour = 82.87, meaning B lineage is notably higher in tumour samples.

MCPcounter & EPIC — both methods independently flag B cells as the only significantly different immune cell type between tumour and normal. That convergence strengthens the finding. Ready to continue.

```
# Checking gene name format in normalized counts
head(rownames(norm_counts), 20)
```

```
##  [1] "CHAC1"   "GLRX"    "MEFV"    "STXBP1"  "AKAP8L"  "AKR1C1"  "CTSL"
##  [8] "IGFBP4"  "COQ8A"   "CD4"     "FDFT1"   "MAPK1"   "STIP1"   "AGRN"
## [15] "ABCG1"   "COPS3"   "PFDN2"   "TSC22D3" "BCL2L2"  "ERBB3"
```

```
# Prepare DESeq2 normalized data for Decosus



# Convert normalized counts to data frame
norm_df_decosus <- as.data.frame(norm_counts)

# Add hgnc_symbols column (required by Decosus)
norm_df_decosus$hgnc_symbols <- rownames(norm_df_decosus)

# Run Decosus consensus deconvolution



est_Decosus <- Decosus::cosDeco(
  x = norm_df_decosus,
  rnaseq = TRUE,
  plot = TRUE
)
```

```
## [1] "Num. of genes: 7849"
```

```
## ℹ GSVA version 2.4.4
```

```
## ℹ Searching for rows with constant values
```

```
## ! Some gene sets have size one. Consider setting minSize > 1
```

```
## ℹ Calculating ssGSEA scores for 486 gene sets
```

```
## ℹ Calculating ranks
```

```
## ℹ Calculating rank weights
```

```
## ✓ Calculations finished
```

```
##
## Setting helper functions for quanTIseq
```

```
##
## Running quanTIseq deconvolution module
```

```
## Gene expression normalization and re-annotation (arrays: FALSE)
```

```
## Removing 17 noisy genes
```

```
## Signature genes found in data set: 104/153 (67.97%)
```

```
## Mixture deconvolution (method: lsei)
```

```
# Extract Results


# Sample-level scores
est_Decosus_samples <- est_Decosus$main_samples

# Cell-type-level scores
est_Decosus_cells <- est_Decosus$main_cells

# Check dimensions
dim(est_Decosus_samples)
```

```
## [1] 80 10
```

```
dim(est_Decosus_cells)
```

```
## [1] 15 10
```

```
# Preview
head(est_Decosus_samples)
```

	S02_Nor... <dbl>	S03_Nor... <dbl>	S05_Nor... <dbl>	S06_Nor. <dl
B cell_consensus	0.5176098	0.2194236	0.5829068	0.41966
Cancer associated fibroblast_consensus	0.4163994	0.1563466	0.3073388	2.82534
cytotoxicity score_consensus	0.6907439	1.0260968	1.0736352	0.72205
Endothelial cell_consensus	0.4793397	0.7841625	1.0318378	1.40662
Macrophage_consensus	0.7381931	1.1080778	1.6261732	0.74190
Macrophage M1_consensus	1.1791018	0.3723611	1.1504008	1.33884

6 rows | 1-6 of 11 columns

```
head(est_Decosus_cells)
```

	S02_Nor... <dbl>	S03_Normal <dbl>	S05_Nor... <dbl>	S06_Nor... <dbl>	S10
B cell_consensus	0.4150400	1.289486e-01	0.6457156	0.4540715	1.
Monocyte_consensus	NaN	NaN	NaN	NaN	
Neutrophil_consensus	0.5349197	1.802626e+00	1.5297952	0.2411260	1.
NK cell_consensus	0.7921439	5.750306e-05	0.2271246	0.9029261	0.
T cell CD8+_consensus	0.5689587	4.888446e-01	0.5062835	1.0206951	1.
uncharacterized cell_consensus	1.1973355	4.248069e-01	0.4612103	0.4799026	0.

6 rows | 1-6 of 11 columns

Decosus ran successfully:

7849 genes used, 80 samples-level scores × 10 samples, 15 cell types × 10 samples 104/153 signature genes found (67.97%) — acceptable coverage Used ssGSEA and quanTIseq internally — confirming it is a consensus method

Results summary:

Cancer associated fibroblasts — S06_Normal extreme outlier again (2.83) vs all others (<0.6), fully consistent with EPIC CAFs and MCP-counter Fibroblasts findings B cell consensus — S05_Tumour (1.42) and S10_Tumour (1.73) notably elevated, again consistent with EPIC and MCP-counter B cell findings

Cytotoxicity score — slightly higher in some tumour samples (S02, S05 Tumour), suggesting modest immune activation

Endothelial cells — S06_Normal elevated (1.41), drops in tumour Macrophage and Macrophage M1 — variable across samples, no clear tumour/normal pattern Macrophage — S05_Normal highest (1.63)

Cross-method consistency: All three methods (EPIC, MCP-counter, Decosus) agree on S06_Normal fibroblast/CAF dominance and B cell elevation in tumour.

```
# Converting Decosus output to numeric matrix
est_Decosus_samples_mat <- as.matrix(est_Decosus_samples)
est_Decosus_samples_mat <- apply(est_Decosus_samples_mat, 2, as.numeric)
rownames(est_Decosus_samples_mat) <- rownames(est_Decosus_samples)

# Identifying fully NaN rows
nan_rows <- apply(est_Decosus_samples_mat, 1, function(x) all(is.nan(x)))
rownames(est_Decosus_samples_mat)[nan_rows]
```

```
## [1] "Monocyte_consensus"
```

```
# Removing fully NaN rows and replacing remaining NaN with 0
est_Decosus_clean <- est_Decosus_samples_mat[!nan_rows, ]
est_Decosus_clean[is.nan(est_Decosus_clean)] <- 0

# Checking cleaned dimensions
dim(est_Decosus_clean)
```

```
## [1] 79 10
```

```
head(est_Decosus_clean)
```

```
##                                       S02_Normal S03_Normal S05_Normal
## B cell_consensus                       0.5176098  0.2194236  0.5829068
## Cancer associated fibroblast_consensus  0.4163994  0.1563466  0.3073388
## cytotoxicity score_consensus           0.6907439  1.0260968  1.0736352
## Endothelial cell_consensus             0.4793397  0.7841625  1.0318378
## Macrophage_consensus                   0.7381931  1.1080778  1.6261732
## Macrophage M1_consensus                1.1791018  0.3723611  1.1504008
##                                       S06_Normal S10_Normal S02_Tumour
## B cell_consensus                       0.4196632  0.7800533  0.4513624
## Cancer associated fibroblast_consensus  2.8253430  0.1726550  0.5764759
## cytotoxicity score_consensus           0.7220515  0.9356024  1.1525859
## Endothelial cell_consensus             1.4066230  0.6790938  0.6227678
## Macrophage_consensus                   0.7419072  0.7835918  1.1087027
## Macrophage M1_consensus                1.3388431  0.4525520  0.5472173
##                                       S03_Tumour S05_Tumour S06_Tumour
## B cell_consensus                       0.5503248 1.41536835  0.6909655
## Cancer associated fibroblast_consensus  0.1443269 0.07552676  0.1087850
## cytotoxicity score_consensus           0.5576894 1.01961531  0.2266804
## Endothelial cell_consensus             0.7288693 0.42606519  0.6439271
## Macrophage_consensus                   0.4437975 0.61069607  0.3831028
## Macrophage M1_consensus                0.2564812 0.44434915  0.5446700
##                                       S10_Tumour
## B cell_consensus                       1.7296703
## Cancer associated fibroblast_consensus  0.1338240
## cytotoxicity score_consensus           1.4248281
## Endothelial cell_consensus             0.4775924
## Macrophage_consensus                   0.4165417
## Macrophage M1_consensus                0.2724281
```

Monocyte_consensus was the only fully NaN row — removed Cleaned matrix: 79 cell types × 10 samples Data looks clean, no remaining NaN issues

Monocyte_consensus being NaN is not unusual — Decosus could not reliably estimate monocyte scores from this dataset, possibly due to insufficient signature gene coverage for that specific cell type.

```
# Checking convergence details for S06_Normal
est_Decosus$fit.gof
```

```
## NULL
```

```
# Checking dimensions of cleaned estimates (cell types x samples)
cat("Dimensions of clean estimates (cell types x samples):\n")
```

```
## Dimensions of clean estimates (cell types x samples):
```

```
dim(est_Decosus_clean)
```

```
## [1] 79 10
```

```
# Checking for any remaining NaN values
cat("\nAny NaN values remaining:\n")
```

```
##
## Any NaN values remaining:
```

```
sum(is.nan(as.matrix(est_Decosus_clean)))
```

```
## [1] 0
```

```
# Checking for any remaining NA values
cat("\nAny NA values remaining:\n")
```

```
##
## Any NA values remaining:
```

```
sum(is.na(as.matrix(est_Decosus_clean)))
```

```
## [1] 0
```

```
# Checking value ranges
cat("\nMin value:\n")
```

```
##
## Min value:
```

```
min(est_Decosus_clean, na.rm = TRUE)
```

```
## [1] 0
```

```
cat("\nMax value:\n")
```

```
##
## Max value:
```

```
max(est_Decosus_clean, na.rm = TRUE)
```

```
## [1] 3
```

```
# Checking column sums per sample
cat("\nColumn sums per sample:\n")
```

```
##
## Column sums per sample:
```

```
colSums(est_Decosus_clean, na.rm = TRUE)
```

```
## S02_Normal S03_Normal S05_Normal S06_Normal S10_Normal S02_Tumour S03_Tumour
##   42.01253   37.20589   47.17014   57.42379   57.88788   40.93572   57.34433
## S05_Tumour S06_Tumour S10_Tumour
##   61.70977   47.11325   51.24757
```

```
# Previewing the full clean estimates matrix
cat("\nFull clean estimates:\n")
```

```
##
## Full clean estimates:
```

```
print(round(est_Decosus_clean, 4))
```

```
##                                          S02_Normal S03_Normal S05_Normal
## B cell_consensus                             0.5176     0.2194     0.5829
## Cancer associated fibroblast_consensus       0.4164     0.1563     0.3073
## cytotoxicity score_consensus                 0.6907     1.0261     1.0736
## Endothelial cell_consensus                   0.4793     0.7842     1.0318
## Macrophage_consensus                         0.7382     1.1081     1.6262
## Macrophage M1_consensus                      1.1791     0.3724     1.1504
## Macrophage M2_consensus                      0.9682     0.8509     1.4044
## Mast cell_consensus                          0.6342     0.7517     0.6008
## Myeloid dendritic cell_consensus             0.3904     0.8324     0.6462
## Neutrophil_consensus                         0.7806     1.4576     0.8717
## NK cell_consensus                            0.4906     0.4779     0.6979
## Plasmacytoid dendritic cell_consensus        0.5726     0.4686     0.9436
## T cell CD4+ (non-regulatory)_consensus       1.4833     0.0000     0.0000
## T cell CD4+ memory_consensus                 0.2792     0.2774     0.6158
## T cell CD4+ Th1_consensus                    0.0000     0.0000     1.7466
## T cell CD4+ Th2_consensus                    0.0000     0.0000     0.0000
## T cell CD8+_consensus                        0.3785     0.4792     0.6042
## T cell CD8+ effector memory_consensus        0.2923     0.4777     0.2873
## T cell regulatory (Tregs)_consensus          0.2663     0.2150     0.3210
## uncharacterized cell_consensus               1.1973     0.4248     0.4612
## aDC_xcell                                    2.7038     1.1579     0.0000
## Adipocytes_xcell                             0.0000     1.9506     1.6891
## Astrocytes_xcell                             0.5129     0.0000     0.0000
## Basophils_xcell                              0.0000     1.3902     0.0000
## CD4_Tcells_EPIC                              1.0334     1.4127     1.5676
## CD4+ naive T-cells_xcell                     0.8842     0.0000     0.0493
## CD4+ Tcm_xcell                               0.0000     0.0000     2.4870
## CD4+ Tem_xcell                               0.0000     0.0000     0.2122
## CD8 effector-NK-cells_Davoli                 0.7942     0.7652     0.7874
## CD8+ naive T-cells_xcell                     0.0000     0.0000     0.8869
## CD8+ Tcm_xcell                               0.0000     0.0000     0.0000
## cDC_xcell                                    0.6288     1.1298     0.9001
## Chondrocytes_xcell                           0.9917     0.6366     0.3028
## Class-switched memory B-cells_xcell          0.0000     0.0000     0.2438
## CLP_xcell                                    2.6206     0.1925     0.8063
## CMP_xcell                                    1.3017     1.0509     0.0000
## Co-inhibition, APC_Rooney                    0.7184     0.8391     1.0307
## Co-inhibition, T cell_Rooney                 0.2829     0.3219     0.6208
## Co-stimulation, APC_Rooney                   1.3041     1.0291     0.7366
## Co-stimulation, T cell_Rooney                0.4973     0.5469     0.6265
## Eosinophils_xcell                            0.0000     0.0000     1.5053
## Epithelial cells_xcell                       0.0000     0.0000     0.0461
## Erythrocytes_xcell                           0.0000     0.0000     0.0000
## Exhausted CD8_Danaher                        0.1896     0.4389     0.8555
## GMP_xcell                                    2.7721     0.0000     0.3929
## Hepatocytes_xcell                            0.0000     0.0000     0.0000
## HSC_xcell                                    1.6292     0.8815     0.5777
## iDC_xcell                                    0.5060     0.3321     0.6947
## ImmuneScore_xcell                            0.1180     0.2879     0.6077
## Keratinocytes_xcell                          0.3493     0.0000     0.0000
## ly Endothelial cells_xcell                   0.0000     0.0000     0.0000
## Megakaryocytes_xcell                         0.0000     0.0000     0.0000
## Melanocytes_xcell                            0.0000     1.3631     1.4728
## Memory B-cells_xcell                         0.0000     0.0000     0.0000
```

```
## MEP_xcell                                0.0000    0.0572    0.0240
## Mesangial cells_xcell                    0.0000    0.0000    0.0000
## MicroenvironmentScore_xcell              0.4789    0.4343    0.9588
## Monocytic lineage_MCP                    0.8299    0.9332    1.3777
## MPP_xcell                                0.0000    0.0000    0.0000
## MSC_xcell                                0.0000    0.5792    0.4100
## mv Endothelial cells_xcell               0.0000    0.0000    2.6653
## Myocytes_xcell                           1.0356    0.0000    0.3617
## naive B-cells_xcell                      0.0000    0.0000    0.0000
## Neurons_xcell                            0.3406    2.0019    0.1916
## NK CD56dim cells_Danaher                 0.2334    0.4634    0.6242
## NKT_xcell                                0.0000    0.8162    0.3803
## Osteoblast_xcell                         0.0000    1.3768    0.0000
## Pericytes_xcell                          0.0000    0.0000    0.0000
## Plasma cells_xcell                       0.0000    0.2363    0.0000
## Platelets_xcell                          1.1586    0.7408    1.4966
## Preadipocytes_xcell                      1.6832    0.6754    1.0765
## pro B-cells_xcell                        0.0000    0.0000    1.1080
## Sebocytes_xcell                          0.0000    0.0000    0.0000
## Skeletal muscle_xcell                    2.0691    0.0000    0.0000
## Smooth muscle_xcell                      1.0289    0.6745    0.4471
## StromaScore_xcell                        0.7170    0.4236    0.9699
## T cells_MCP                              0.4761    0.6694    0.5753
## T-cells_Danaher                          0.3679    1.0170    0.4302
## Tgd cells_xcell                          0.0000    0.0000    0.0000
##                                      S06_Normal S10_Normal S02_Tumour
## B cell_consensus                         0.4197    0.7801    0.4514
## Cancer associated fibroblast_consensus   2.8253    0.1727    0.5765
## cytotoxicity score_consensus             0.7221    0.9356    1.1526
## Endothelial cell_consensus               1.4066    0.6791    0.6228
## Macrophage_consensus                     0.7419    0.7836    1.1087
## Macrophage M1_consensus                  1.3388    0.4526    0.5472
## Macrophage M2_consensus                  0.4998    0.7709    0.7815
## Mast cell_consensus                      1.7230    1.5608    0.9485
## Myeloid dendritic cell_consensus         0.7313    1.4627    0.3547
## Neutrophil_consensus                     0.5393    0.7355    0.7841
## NK cell_consensus                        0.8283    0.5135    0.8808
## Plasmacytoid dendritic cell_consensus    0.3176    0.3372    0.7434
## T cell CD4+ (non-regulatory)_consensus   0.0000    0.2230    0.0000
## T cell CD4+ memory_consensus             0.9689    0.4495    1.0469
## T cell CD4+ Th1_consensus                0.0000    0.0000    0.0000
## T cell CD4+ Th2_consensus                1.9913    0.5379    1.0344
## T cell CD8+_consensus                    0.8723    1.1813    0.7962
## T cell CD8+ effector memory_consensus    0.3798    0.9222    1.1685
## T cell regulatory (Tregs)_consensus      0.2828    0.4493    0.6574
## uncharacterized cell_consensus           0.4799    0.6871    0.8994
## aDC_xcell                                0.0437    0.0000    0.5824
## Adipocytes_xcell                         0.4605    1.4595    0.0000
## Astrocytes_xcell                         0.0000    0.0000    0.0000
## Basophils_xcell                          0.0000    0.0000    0.0000
## CD4_Tcells_EPIC                          0.0000    1.1859    0.8132
## CD4+ naive T-cells_xcell                 0.0000    2.4214    1.0812
## CD4+ Tcm_xcell                           0.0000    0.0000    0.0000
## CD4+ Tem_xcell                           0.0000    0.5965    0.6526
## CD8 effector-NK-cells_Davoli             1.0540    1.0380    1.1817
## CD8+ naive T-cells_xcell                 0.0000    0.0000    0.0000
```

```
## CD8+ Tcm_xcell                        0.0000     1.5361     0.7800
## cDC_xcell                             0.6889     2.4150     0.0814
## Chondrocytes_xcell                    2.4570     0.9693     0.7370
## Class-switched memory B-cells_xcell   0.0000     0.0000     0.4157
## CLP_xcell                             0.0873     0.5452     0.7325
## CMP_xcell                             1.8682     0.5233     0.0000
## Co-inhibition, APC_Rooney             0.6973     1.0414     1.3972
## Co-inhibition, T cell_Rooney          0.2632     0.6072     0.7433
## Co-stimulation, APC_Rooney            0.8079     1.3171     1.2815
## Co-stimulation, T cell_Rooney         0.3814     0.8120     0.7278
## Eosinophils_xcell                     0.9656     2.4086     0.0000
## Epithelial cells_xcell                0.0000     0.0000     0.5023
## Erythrocytes_xcell                    0.0000     0.0000     0.0000
## Exhausted CD8_Danaher                 0.2397     0.6004     0.6202
## GMP_xcell                             0.0000     0.4014     0.2293
## Hepatocytes_xcell                     0.7951     2.8927     0.0000
## HSC_xcell                             1.6919     0.8974     1.0125
## iDC_xcell                             0.2421     2.6737     0.0000
## ImmuneScore_xcell                     0.4057     1.3499     0.5335
## Keratinocytes_xcell                   0.0000     0.0000     1.4472
## ly Endothelial cells_xcell            0.8954     0.0000     0.0000
## Megakaryocytes_xcell                  3.0000     0.0000     0.0000
## Melanocytes_xcell                     1.9716     0.0000     0.7388
## Memory B-cells_xcell                  0.0000     0.2443     0.0000
## MEP_xcell                             0.5833     0.7418     0.0000
## Mesangial cells_xcell                 3.0000     0.0000     0.0000
## MicroenvironmentScore_xcell           1.7662     1.1315     0.4965
## Monocytic lineage_MCP                 0.7890     0.8474     1.1202
## MPP_xcell                             0.0000     0.0000     0.0000
## MSC_xcell                             0.9914     0.0000     1.0858
## mv Endothelial cells_xcell            1.3771     0.0000     0.0000
## Myocytes_xcell                        2.3918     0.4092     0.3971
## naive B-cells_xcell                   0.0000     0.0000     0.0000
## Neurons_xcell                         2.0469     0.7954     0.0273
## NK CD56dim cells_Danaher              0.2710     0.3428     0.5321
## NKT_xcell                             1.2151     0.9417     0.1417
## Osteoblast_xcell                      0.0000     0.6543     0.0000
## Pericytes_xcell                       0.1736     0.3762     1.1691
## Plasma cells_xcell                    0.0000     1.2596     0.0000
## Platelets_xcell                       0.0000     1.4964     0.6613
## Preadipocytes_xcell                   1.6151     1.1722     0.7023
## pro B-cells_xcell                     0.0000     0.0000     0.0000
## Sebocytes_xcell                       0.0000     0.0000     0.0000
## Skeletal muscle_xcell                 0.0054     1.6088     0.0000
## Smooth muscle_xcell                   1.2017     1.3298     1.4343
## StromaScore_xcell                     2.6814     0.3516     0.2279
## T cells_MCP                           0.5151     1.1635     0.9905
## T-cells_Danaher                       0.7144     1.6947     1.1034
## Tgd cells_xcell                       0.0000     0.0000     0.0000
##                                     S03_Tumour S05_Tumour S06_Tumour
## B cell_consensus                      0.5503     1.4154     0.6910
## Cancer associated fibroblast_consensus     0.1443     0.0755     0.1088
## cytotoxicity score_consensus          0.5577     1.0196     0.2267
## Endothelial cell_consensus            0.7289     0.4261     0.6439
## Macrophage_consensus                  0.4438     0.6107     0.3831
## Macrophage M1_consensus               0.2565     0.4443     0.5447
```

```
## Macrophage M2_consensus                   0.5945     0.5981     0.6321
## Mast cell_consensus                       0.6012     0.3250     0.4622
## Myeloid dendritic cell_consensus          0.4165     0.6599     0.7049
## Neutrophil_consensus                      0.5363     0.7247     0.5544
## NK cell_consensus                         0.2096     1.5265     0.2122
## Plasmacytoid dendritic cell_consensus     0.6045     0.7284     1.3212
## T cell CD4+ (non-regulatory)_consensus    1.0849     1.0358     0.0000
## T cell CD4+ memory_consensus              1.1926     1.4459     0.3653
## T cell CD4+ Th1_consensus                 1.4706     0.5334     1.8714
## T cell CD4+ Th2_consensus                 0.0755     0.8637     0.1802
## T cell CD8+_consensus                     0.4399     1.4364     0.2073
## T cell CD8+ effector memory_consensus     1.0930     1.4892     0.0992
## T cell regulatory (Tregs)_consensus       1.0236     1.8002     1.5191
## uncharacterized cell_consensus            1.0031     1.0206     1.0289
## aDC_xcell                                 0.0000     0.0874     0.0000
## Adipocytes_xcell                          0.0000     0.0000     0.0000
## Astrocytes_xcell                          2.7134     0.6820     0.0000
## Basophils_xcell                           0.0000     0.0000     2.6584
## CD4_Tcells_EPIC                           0.8924     0.3173     0.1026
## CD4+ naive T-cells_xcell                  0.1842     0.4277     0.0000
## CD4+ Tcm_xcell                            1.6673     0.1865     0.0000
## CD4+ Tem_xcell                            2.2193     1.4940     1.0080
## CD8 effector-NK-cells_Davoli              0.8798     1.1082     0.7500
## CD8+ naive T-cells_xcell                  1.9918     0.6167     0.0000
## CD8+ Tcm_xcell                            1.3727     1.5349     0.0000
## cDC_xcell                                 0.0960     0.4032     0.0000
## Chondrocytes_xcell                        0.0000     0.0000     0.0000
## Class-switched memory B-cells_xcell       0.6871     2.0931     1.9476
## CLP_xcell                                 0.1615     0.7601     0.0000
## CMP_xcell                                 1.5611     0.0000     0.0000
## Co-inhibition, APC_Rooney                 0.4440     1.5551     0.0000
## Co-inhibition, T cell_Rooney              1.2111     1.8735     1.0963
## Co-stimulation, APC_Rooney                0.5123     0.4883     0.0000
## Co-stimulation, T cell_Rooney             1.5345     1.5935     0.9866
## Eosinophils_xcell                         0.0000     0.0089     0.0000
## Epithelial cells_xcell                    1.2384     1.7513     2.0166
## Erythrocytes_xcell                        1.1257     2.0745     1.8518
## Exhausted CD8_Danaher                     1.1902     1.2617     2.0179
## GMP_xcell                                 0.0000     0.8425     0.0000
## Hepatocytes_xcell                         0.0000     0.0000     0.0000
## HSC_xcell                                 0.2108     0.0000     0.0000
## iDC_xcell                                 0.0000     0.6609     0.0000
## ImmuneScore_xcell                         0.5940     1.5541     0.1565
## Keratinocytes_xcell                       1.4310     2.1560     0.0000
## ly Endothelial cells_xcell                2.5354     0.0000     1.3304
## Megakaryocytes_xcell                      0.0000     0.0000     0.0000
## Melanocytes_xcell                         0.1742     0.3050     0.6453
## Memory B-cells_xcell                      0.8753     1.3556     0.0000
## MEP_xcell                                 0.5528     1.0745     0.0461
## Mesangial cells_xcell                     0.0000     0.0000     0.0000
## MicroenvironmentScore_xcell               0.4122     1.0785     0.1086
## Monocytic lineage_MCP                     0.8697     0.9244     1.0575
## MPP_xcell                                 0.0000     0.0000     3.0000
## MSC_xcell                                 2.2038     0.8550     0.7189
## mv Endothelial cells_xcell                0.0000     0.0000     0.0000
## Myocytes_xcell                            0.0000     0.2675     1.2958
```

```
## naive B-cells_xcell                          0.0052     2.3354     0.0713
## Neurons_xcell                                0.1293     0.0000     0.0000
## NK CD56dim cells_Danaher                     1.2218     2.1256     1.0991
## NKT_xcell                                    0.2872     0.2394     2.3803
## Osteoblast_xcell                             0.0000     0.0000     2.5773
## Pericytes_xcell                              2.1140     1.4165     0.9929
## Plasma cells_xcell                           1.5564     0.0000     0.0000
## Platelets_xcell                              0.6818     0.2309     0.0000
## Preadipocytes_xcell                          0.0000     0.0000     0.0000
## pro B-cells_xcell                            0.3507     0.6015     0.7206
## Sebocytes_xcell                              1.0569     2.2207     1.7180
## Skeletal muscle_xcell                        0.0000     0.3747     1.2346
## Smooth muscle_xcell                          0.1680     0.5225     0.0000
## StromaScore_xcell                            0.0000     0.0000     0.0000
## T cells_MCP                                  1.2128     1.3702     1.0804
## T-cells_Danaher                              0.9908     0.7012     0.6874
## Tgd cells_xcell                              3.0000     0.0000     0.0000
##                                          S10_Tumour
## B cell_consensus                             1.7297
## Cancer associated fibroblast_consensus       0.1338
## cytotoxicity score_consensus                 1.4248
## Endothelial cell_consensus                   0.4776
## Macrophage_consensus                         0.4165
## Macrophage M1_consensus                      0.2724
## Macrophage M2_consensus                      0.6995
## Mast cell_consensus                          0.4189
## Myeloid dendritic cell_consensus             0.4557
## Neutrophil_consensus                         0.4931
## NK cell_consensus                            0.5934
## Plasmacytoid dendritic cell_consensus        1.6288
## T cell CD4+ (non-regulatory)_consensus       0.0000
## T cell CD4+ memory_consensus                 0.2403
## T cell CD4+ Th1_consensus                    0.0000
## T cell CD4+ Th2_consensus                    1.7003
## T cell CD8+_consensus                        1.2352
## T cell CD8+ effector memory_consensus        0.5469
## T cell regulatory (Tregs)_consensus          0.7716
## uncharacterized cell_consensus               1.3158
## aDC_xcell                                    0.0000
## Adipocytes_xcell                             0.0000
## Astrocytes_xcell                             0.9535
## Basophils_xcell                              0.0000
## CD4_Tcells_EPIC                              0.7100
## CD4+ naive T-cells_xcell                     0.9832
## CD4+ Tcm_xcell                               0.0000
## CD4+ Tem_xcell                               0.0000
## CD8 effector-NK-cells_Davoli                 1.0071
## CD8+ naive T-cells_xcell                     1.9662
## CD8+ Tcm_xcell                               1.3386
## cDC_xcell                                    0.1809
## Chondrocytes_xcell                           0.0000
## Class-switched memory B-cells_xcell          0.3485
## CLP_xcell                                    0.0000
## CMP_xcell                                    0.0000
## Co-inhibition, APC_Rooney                    0.7610
## Co-inhibition, T cell_Rooney                 1.1232
```

```
## Co-stimulation, APC_Rooney            1.0803
## Co-stimulation, T cell_Rooney         0.9270
## Eosinophils_xcell                     0.0000
## Epithelial cells_xcell                0.2792
## Erythrocytes_xcell                    0.0000
## Exhausted CD8_Danaher                 0.3953
## GMP_xcell                             0.4875
## Hepatocytes_xcell                     0.0000
## HSC_xcell                             0.7053
## iDC_xcell                             0.7120
## ImmuneScore_xcell                     1.8628
## Keratinocytes_xcell                   0.2955
## ly Endothelial cells_xcell            0.0000
## Megakaryocytes_xcell                  0.0000
## Melanocytes_xcell                     0.0000
## Memory B-cells_xcell                  2.5173
## MEP_xcell                             2.5775
## Mesangial cells_xcell                 0.0000
## MicroenvironmentScore_xcell           1.2928
## Monocytic lineage_MCP                 0.4664
## MPP_xcell                             0.0000
## MSC_xcell                             0.4795
## mv Endothelial cells_xcell            0.0000
## Myocytes_xcell                        0.0000
## naive B-cells_xcell                   1.8817
## Neurons_xcell                         0.0000
## NK CD56dim cells_Danaher              0.8050
## NKT_xcell                             0.0000
## Osteoblast_xcell                      0.1846
## Pericytes_xcell                       0.0000
## Plasma cells_xcell                    2.2215
## Platelets_xcell                       1.2939
## Preadipocytes_xcell                   0.2759
## pro B-cells_xcell                     2.6016
## Sebocytes_xcell                       0.0000
## Skeletal muscle_xcell                 0.6825
## Smooth muscle_xcell                   1.3099
## StromaScore_xcell                     0.0000
## T cells_MCP                           0.9372
## T-cells_Danaher                       1.0486
## Tgd cells_xcell                       0.0000
```

Data is clean, ready for heatmap and paired t-test.

```
# Converting Decosus clean matrix to data frame
decosus_df <- as.data.frame(est_Decosus_clean)

# Extracting group and patient ID from column names
sample_names <- colnames(decosus_df)
group <- ifelse(grepl("Tumour", sample_names), "Tumour", "Normal")
patient <- gsub("_(Tumour|Normal)", "", sample_names)

# Initialising results storage
decosus_ttest_results <- data.frame(
  Cell_Type = rownames(decosus_df),
  Mean_Normal = NA,
  Mean_Tumour = NA,
  P_value = NA
)

# Looping through each cell type
for (i in 1:nrow(decosus_df)) {

  values <- as.numeric(decosus_df[i, ])

  # Building data frame for patient-matched ordering
  df_temp <- data.frame(value = values, group = group, patient = patient)

  # Ordering normal and tumour values by patient ID
  normal_ordered <- df_temp[df_temp$group == "Normal", ]
  tumour_ordered <- df_temp[df_temp$group == "Tumour", ]
  normal_ordered <- normal_ordered[order(normal_ordered$patient), ]
  tumour_ordered <- tumour_ordered[order(tumour_ordered$patient), ]

  # Performing paired t-test
  test <- t.test(tumour_ordered$value, normal_ordered$value, paired = TRUE)

  # Storing mean values and p-value
  decosus_ttest_results$Mean_Normal[i] <- mean(normal_ordered$value)
  decosus_ttest_results$Mean_Tumour[i] <- mean(tumour_ordered$value)
  decosus_ttest_results$P_value[i] <- test$p.value
}

# Viewing all results
decosus_ttest_results
```

Cell_Type <chr>	Mean_Normal <dbl>	Mean_Tumour <dbl>	P_value <dbl>
B cell_consensus	5.039314e-01	9.675383e-01	0.069336997
Cancer associated fibroblast_consensus	7.756166e-01	2.077877e-01	0.352948934
cytotoxicity score_consensus	8.896260e-01	8.762798e-01	0.953353981
Endothelial cell_consensus	8.762114e-01	5.798444e-01	0.155050540
Macrophage_consensus	9.995886e-01	5.925682e-01	0.149556679
Macrophage M1_consensus	8.986517e-01	4.130292e-01	0.025946327

Cell_Type <chr>	Mean_Normal <dbl>	Mean_Tumour <dbl>	P_value <dbl>
Macrophage M2_consensus	8.988455e-01	6.611455e-01	0.203780197
Mast cell_consensus	1.054115e+00	5.511719e-01	0.171381537
Myeloid dendritic cell_consensus	8.125858e-01	5.183429e-01	0.204784988
Neutrophil_consensus	8.769116e-01	6.185220e-01	0.208638190

1-10 of 79 rows Previous 1 2 3 4 5 6 ... 8 Next

```
# Identifying significant cell types (p < 0.05)
decosus_significant <- decosus_ttest_results[decosus_ttest_results$P_value < 0.05, ]
decosus_significant
```

	Cell_Type <chr>	Mean_Normal <dbl>	Mean_Tumour <dbl>	P_value <dbl>
6	Macrophage M1_consensus	0.898651738	4.130292e-01	0.025946327
19	T cell regulatory (Tregs)_consensus	0.306884488	1.154358e+00	0.020404847
22	Adipocytes_xcell	1.111937985	2.827759e-17	0.041460137
32	cDC_xcell	1.152516446	1.523005e-01	0.036143363
34	Class-switched memory B-cells_xcell	0.048763832	1.098399e+00	0.040481977
38	Co-inhibition, T cell_Rooney	0.419216286	1.209499e+00	0.005247283
40	Co-stimulation, T cell_Rooney	0.572806527	1.153882e+00	0.032554596
42	Epithelial cells_xcell	0.009225988	1.157577e+00	0.026593422
47	HSC_xcell	1.135540120	3.857205e-01	0.040080017
65	NK CD56dim cells_Danaher	0.386948069	1.156725e+00	0.020436749

1-10 of 11 rows Previous 1 2 Next

```
# Viewing all results
decosus_ttest_results
```

Cell_Type <chr>	Mean_Normal <dbl>	Mean_Tumour <dbl>	P_value <dbl>
B cell_consensus	5.039314e-01	9.675383e-01	0.069336997
Cancer associated fibroblast_consensus	7.756166e-01	2.077877e-01	0.352948934
cytotoxicity score_consensus	8.896260e-01	8.762798e-01	0.953353981
Endothelial cell_consensus	8.762114e-01	5.798444e-01	0.155050540
Macrophage_consensus	9.995886e-01	5.925682e-01	0.149556679
Macrophage M1_consensus	8.986517e-01	4.130292e-01	0.025946327
Macrophage M2_consensus	8.988455e-01	6.611455e-01	0.203780197

Cell_Type <chr>	Mean_Normal <dbl>	Mean_Tumour <dbl>	P_value <dbl>
Mast cell_consensus	1.054115e+00	5.511719e-01	0.171381537
Myeloid dendritic cell_consensus	8.125858e-01	5.183429e-01	0.204784988
Neutrophil_consensus	8.769116e-01	6.185220e-01	0.208638190

1-10 of 79 rows Previous **1** 2 3 4 5 6 ... 8 Next

```
# Identifying and displaying significant cell types (p < 0.05)
decosus_significant <- decosus_ttest_results[decosus_ttest_results$P_value < 0.05, ]

# Sorting by p-value ascending
decosus_significant <- decosus_significant[order(decosus_significant$P_value), ]
rownames(decosus_significant) <- NULL

cat("Significant cell types (p < 0.05):\n")
```

```
## Significant cell types (p < 0.05):
```

```
decosus_significant
```

Cell_Type <chr>	Mean_Normal <dbl>	Mean_Tumour <dbl>	P_value <dbl>
Preadipocytes_xcell	1.244468709	1.956335e-01	0.002569480
Co-inhibition, T cell_Rooney	0.419216286	1.209499e+00	0.005247283
T cell regulatory (Tregs)_consensus	0.306884488	1.154358e+00	0.020404847
NK CD56dim cells_Danaher	0.386948069	1.156725e+00	0.020436749
Macrophage M1_consensus	0.898651738	4.130292e-01	0.025946327
Epithelial cells_xcell	0.009225988	1.157577e+00	0.026593422
Co-stimulation, T cell_Rooney	0.572806527	1.153882e+00	0.032554596
cDC_xcell	1.152516446	1.523005e-01	0.036143363
HSC_xcell	1.135540120	3.857205e-01	0.040080017
Class-switched memory B-cells_xcell	0.048763832	1.098399e+00	0.040481977

1-10 of 11 rows Previous **1** 2 Next

```
library(ggplot2)
library(tidyr)
```

```
##
## Attaching package: 'tidyr'
```

```
## The following object is masked from 'package:S4Vectors':
##
##     expand
```

```
library(dplyr)

today <- format(Sys.Date(), "%Y-%m-%d")

save_path <- "/Users/dr.parul/Downloads/data_required_for_CAN7032-7132/LAQ2_data/"

# ===============================
# Create group vector dynamically from column names
# ===============================

create_group_vector <- function(est_matrix) {
  sample_names <- colnames(est_matrix)
  ifelse(grepl("Tumour", sample_names), "Tumour", "Normal")
}

# ===============================
# Run t-tests
# ===============================

run_ttests <- function(est_matrix) {

  est_matrix <- as.matrix(est_matrix)
  group <- create_group_vector(est_matrix)

  pvals <- apply(est_matrix, 1, function(x) {
    tryCatch(
      t.test(x[group == "Normal"], x[group == "Tumour"])$p.value,
      error = function(e) NA
    )
  })

  sig <- pvals[!is.na(pvals) & pvals < 0.05]
  return(sig)
}

# ===============================
# Plot boxplots for significant cell types
# ===============================

plot_boxplots <- function(est_matrix, sig_pvals, method_name, save_path) {

  if (length(sig_pvals) == 0) {
    cat(method_name, ": No significant cell types found (p < 0.05)\n")
    return(invisible(NULL))
  }

  est_matrix <- as.matrix(est_matrix)
  group <- create_group_vector(est_matrix)

  sig_cells <- names(sig_pvals)

  sig_data <- as.data.frame(t(est_matrix[sig_cells, , drop = FALSE]))
  sig_data$Group <- group

  long_data <- pivot_longer(
```

```
    sig_data,
    -Group,
    names_to = "CellType",
    values_to = "Estimate"
  )

  p <- ggplot(long_data, aes(x = Group, y = Estimate, fill = Group)) +
    geom_boxplot(outlier.shape = 16, outlier.size = 2) +
    facet_wrap(~ CellType, scales = "free_y") +
    scale_fill_manual(values = c(Normal = "steelblue", Tumour = "firebrick")) +
    theme_bw() +
    theme(
      strip.text = element_text(size = 8),
      axis.text.x = element_text(angle = 45, hjust = 1)
    ) +
    labs(
      title = paste(method_name, "- Significant Cell Types (p < 0.05)"),
      y = "Cell Abundance Estimate",
      x = "Group"
    )

  ggsave(
    paste0(save_path,
           tolower(gsub(" ", "_", method_name)),
           "_boxplots_",
           today,
           ".png"),
    plot = p,
    width = 10,
    height = 8,
    dpi = 150
  )

  print(p)
}

# ===============================
# Run for all three methods
# ===============================

sig_Decosus    <- run_ttests(est_Decosus_clean)
sig_EPIC       <- run_ttests(est_EPIC_cellProp)
sig_MCPcounter <- run_ttests(est_MCPcounter)

cat("DECOSUS significant cell types:\n"); print(sig_Decosus)
```

```
## DECOSUS significant cell types:
```

```
## T cell regulatory (Tregs)_consensus      uncharacterized cell_consensus
##                          0.016931884                        0.046931213
##                      Adipocytes_xcell                          cDC_xcell
##                          0.041460137                        0.036344236
##          Co-inhibition, T cell_Rooney       Co-stimulation, T cell_Rooney
##                          0.009251208                        0.024592956
##                Epithelial cells_xcell                          HSC_xcell
##                          0.027554419                        0.037557892
##              NK CD56dim cells_Danaher                    Pericytes_xcell
##                          0.044352840                        0.038185028
##                   Preadipocytes_xcell                        T cells_MCP
##                          0.002319963                        0.021926639
```

```
cat("EPIC significant cell types:\n"); print(sig_EPIC)
```

```
## EPIC significant cell types:
```

```
## named numeric(0)
```

```
cat("MCP-counter significant cell types:\n"); print(sig_MCPcounter)
```

```
## MCP-counter significant cell types:
```

```
##    T cells
## 0.02192664
```

```
cat("DECOSUS sig:", length(sig_Decosus), "\n")
```

```
## DECOSUS sig: 12
```

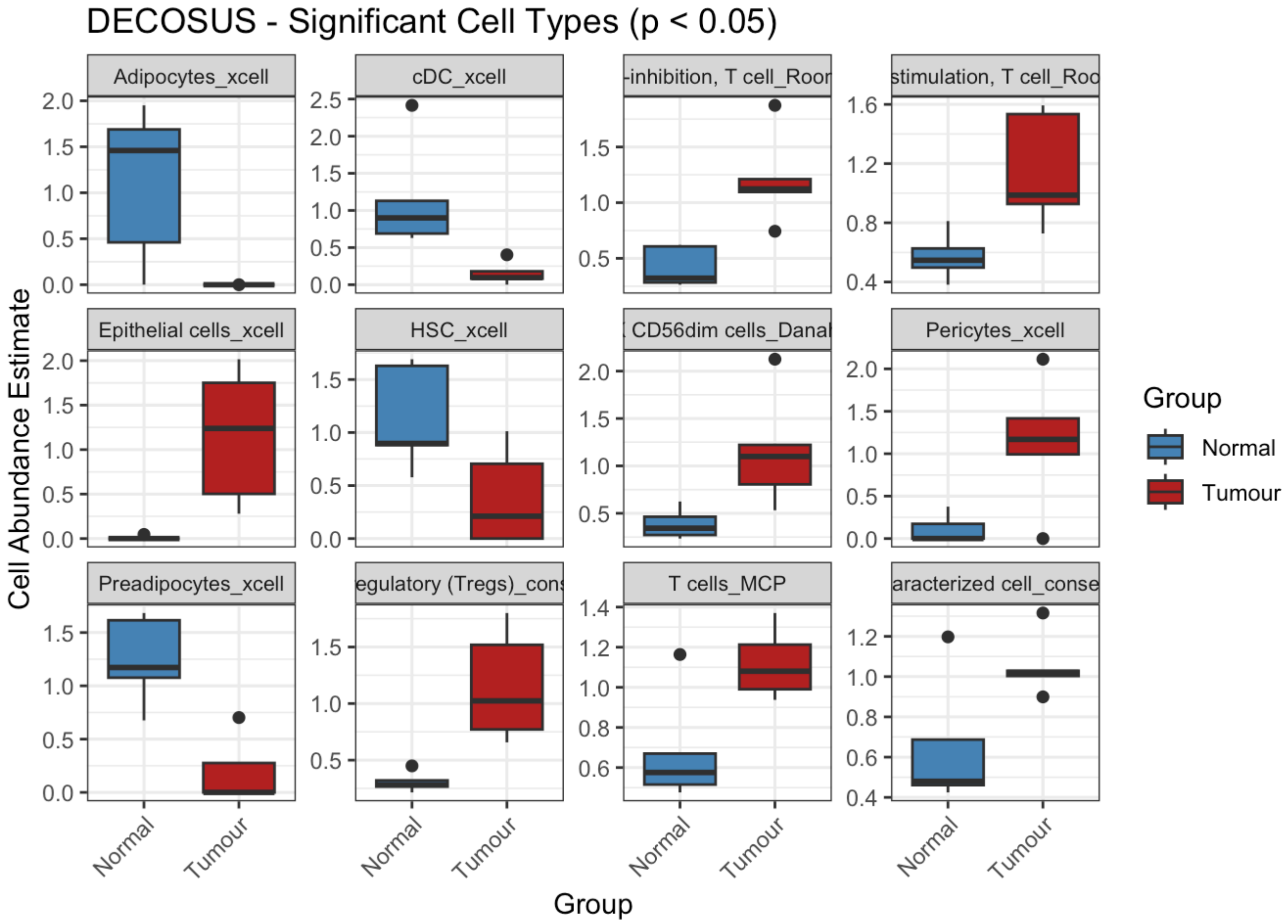
```
cat("EPIC sig:", length(sig_EPIC), "\n")
```

```
## EPIC sig: 0
```

```
cat("MCPcounter sig:", length(sig_MCPcounter), "\n")
```

```
## MCPcounter sig: 1
```

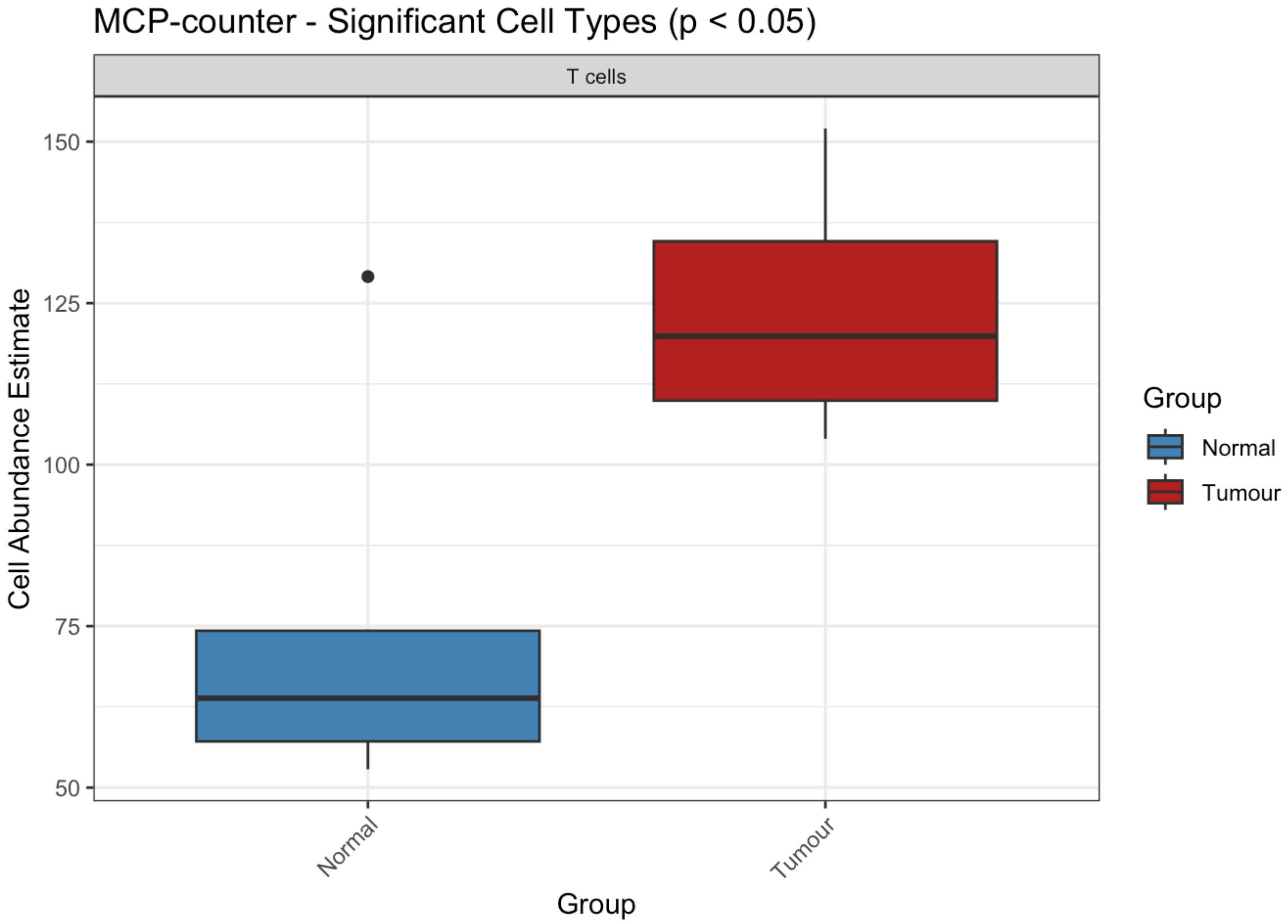
```
plot_boxplots(est_Decosus_clean, sig_Decosus, "DECOSUS", save_path)
```




```
plot_boxplots(est_EPIC_cellProp, sig_EPIC, "EPIC", save_path)
```

```
## EPIC : No significant cell types found (p < 0.05)
```

```
plot_boxplots(est_MCPcounter, sig_MCPcounter, "MCP-counter", save_path)
```


MCP-counter - Significant Cell Types (p < 0.05)
T cells
Cell Abundance Estimate
150
125
100
75
50
Normal
Tumour
Group
Group
Normal
Tumour

```
library(ggplot2)
library(tidyr)
library(dplyr)

save_path <- "/Users/dr.parul/Downloads/data_required_for_CAN7032-7132/LAQ2_data/"

# Creating group vector from column names
create_group_vector <- function(est_matrix) {
  sample_names <- colnames(est_matrix)
  ifelse(grepl("Tumour", sample_names), "Tumour", "Normal")
}

# Extracting patient IDs from column names
create_patient_vector <- function(est_matrix) {
  gsub("_(Tumour|Normal)", "", colnames(est_matrix))
}

# Running paired t-tests per cell type
run_ttests <- function(est_matrix) {

  est_matrix <- as.matrix(est_matrix)
  group <- create_group_vector(est_matrix)
  patient <- create_patient_vector(est_matrix)

  pvals <- apply(est_matrix, 1, function(x) {
    tryCatch({
      df_temp <- data.frame(value = x, group = group, patient = patient)
      normal_ordered <- df_temp[df_temp$group == "Normal", ]
      tumour_ordered <- df_temp[df_temp$group == "Tumour", ]
      normal_ordered <- normal_ordered[order(normal_ordered$patient), ]
      tumour_ordered <- tumour_ordered[order(tumour_ordered$patient), ]
      t.test(tumour_ordered$value, normal_ordered$value, paired = TRUE)$p.value
    }, error = function(e) NA)
  })

  sig <- pvals[!is.na(pvals) & pvals < 0.05]
  return(sig)
}

# Plotting boxplots for significant cell types
plot_boxplots <- function(est_matrix, sig_pvals, method_name, save_path) {

  if (length(sig_pvals) == 0) {
    cat(method_name, ": No significant cell types found (p < 0.05)\n")
    return(invisible(NULL))
  }

  est_matrix <- as.matrix(est_matrix)
  group <- create_group_vector(est_matrix)

  sig_cells <- names(sig_pvals)

  sig_data <- as.data.frame(t(est_matrix[sig_cells, , drop = FALSE]))
  sig_data$Group <- group
```

```
  long_data <- pivot_longer(
    sig_data,
    -Group,
    names_to = "CellType",
    values_to = "Estimate"
  )

  p <- ggplot(long_data, aes(x = Group, y = Estimate, fill = Group)) +
    geom_boxplot(outlier.shape = 16, outlier.size = 2) +
    facet_wrap(~ CellType, scales = "free_y") +
    scale_fill_manual(values = c(Normal = "steelblue", Tumour = "firebrick")) +
    theme_bw() +
    theme(
      strip.text = element_text(size = 8),
      axis.text.x = element_text(angle = 45, hjust = 1)
    ) +
    labs(
      title = paste(method_name, "- Significant Cell Types (p < 0.05)"),
      y = "Cell Abundance Estimate",
      x = "Group"
    )

  ggsave(
    paste0(save_path,
           tolower(gsub(" ", "_", method_name)),
           "_boxplots_2025-02-25.png"),
    plot = p,
    width = 10,
    height = 8,
    dpi = 150
  )

  print(p)
}

# Running for all three methods
sig_Decosus    <- run_ttests(est_Decosus_clean)
sig_EPIC       <- run_ttests(est_EPIC_cellProp)
sig_MCPcounter <- run_ttests(est_MCPcounter)

cat("DECOSUS significant cell types:\n"); print(sig_Decosus)
```

```
## DECOSUS significant cell types:
```

```
##             Macrophage M1_consensus T cell regulatory (Tregs)_consensus
##                         0.025946327                         0.020404847
##                    Adipocytes_xcell                           cDC_xcell
##                         0.041460137                         0.036143363
## Class-switched memory B-cells_xcell        Co-inhibition, T cell_Rooney
##                         0.040481977                         0.005247283
##       Co-stimulation, T cell_Rooney             Epithelial cells_xcell
##                         0.032554596                         0.026593422
##                           HSC_xcell             NK CD56dim cells_Danaher
##                         0.040080017                         0.020436749
##                 Preadipocytes_xcell
##                         0.002569480
```

```
cat("EPIC significant cell types:\n"); print(sig_EPIC)
```

```
## EPIC significant cell types:
```

```
##     Bcells
## 0.04541982
```

```
cat("MCP-counter significant cell types:\n"); print(sig_MCPcounter)
```

```
## MCP-counter significant cell types:
```

```
## B lineage
## 0.0368039
```

```
cat("DECOSUS sig:", length(sig_Decosus), "\n")
```

```
## DECOSUS sig: 11
```

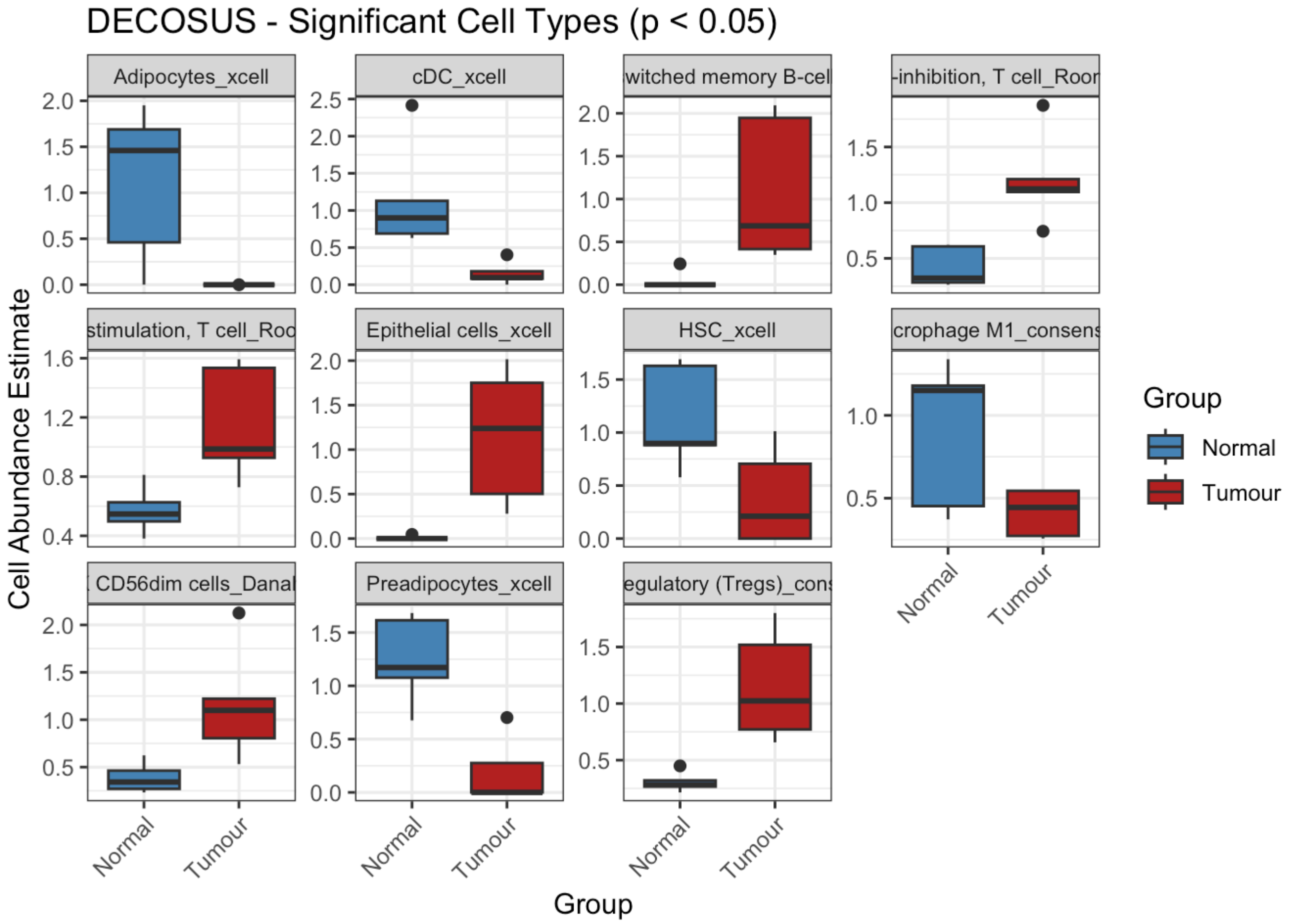
```
cat("EPIC sig:", length(sig_EPIC), "\n")
```

```
## EPIC sig: 1
```

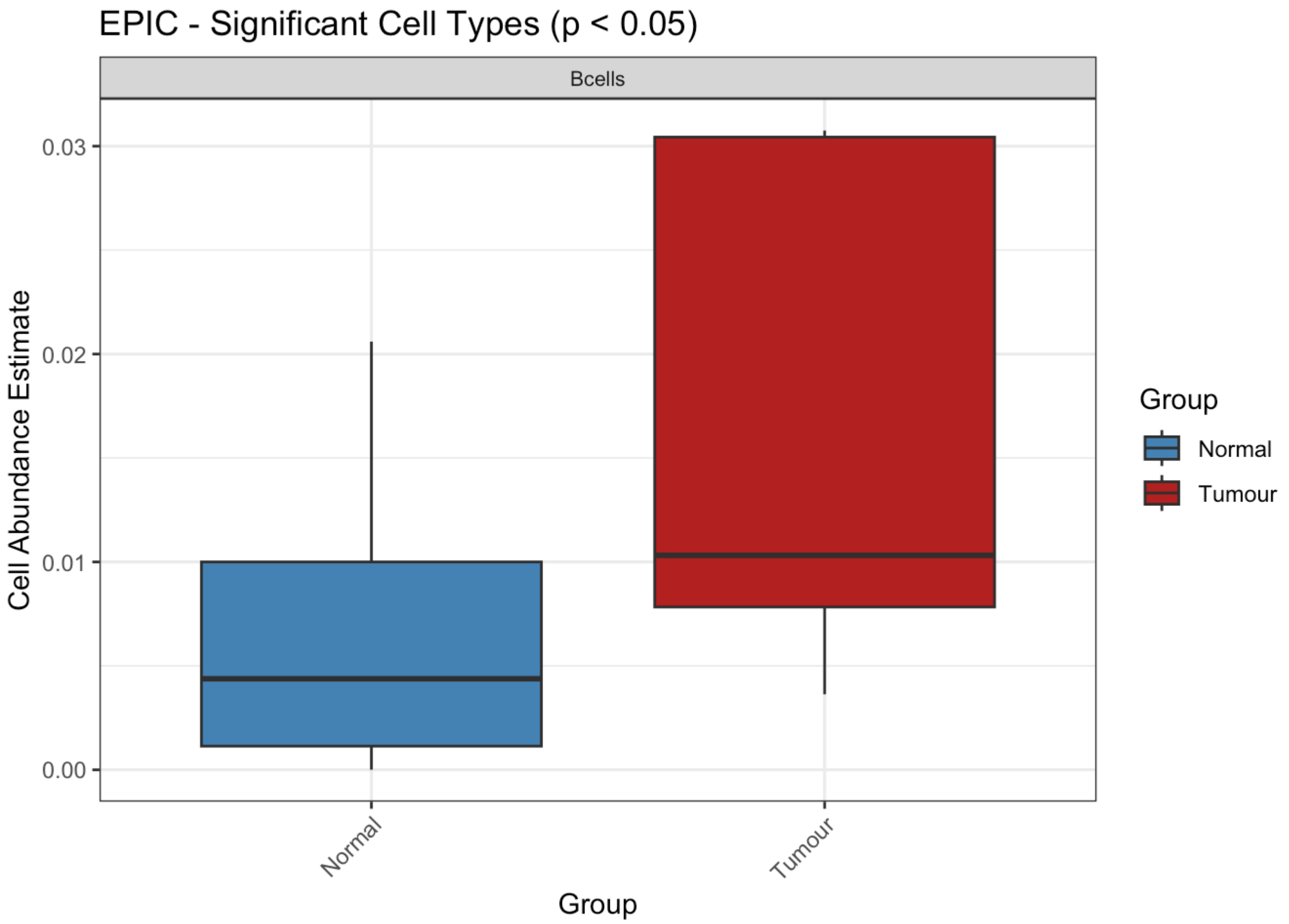
```
cat("MCPcounter sig:", length(sig_MCPcounter), "\n")
```

```
## MCPcounter sig: 1
```

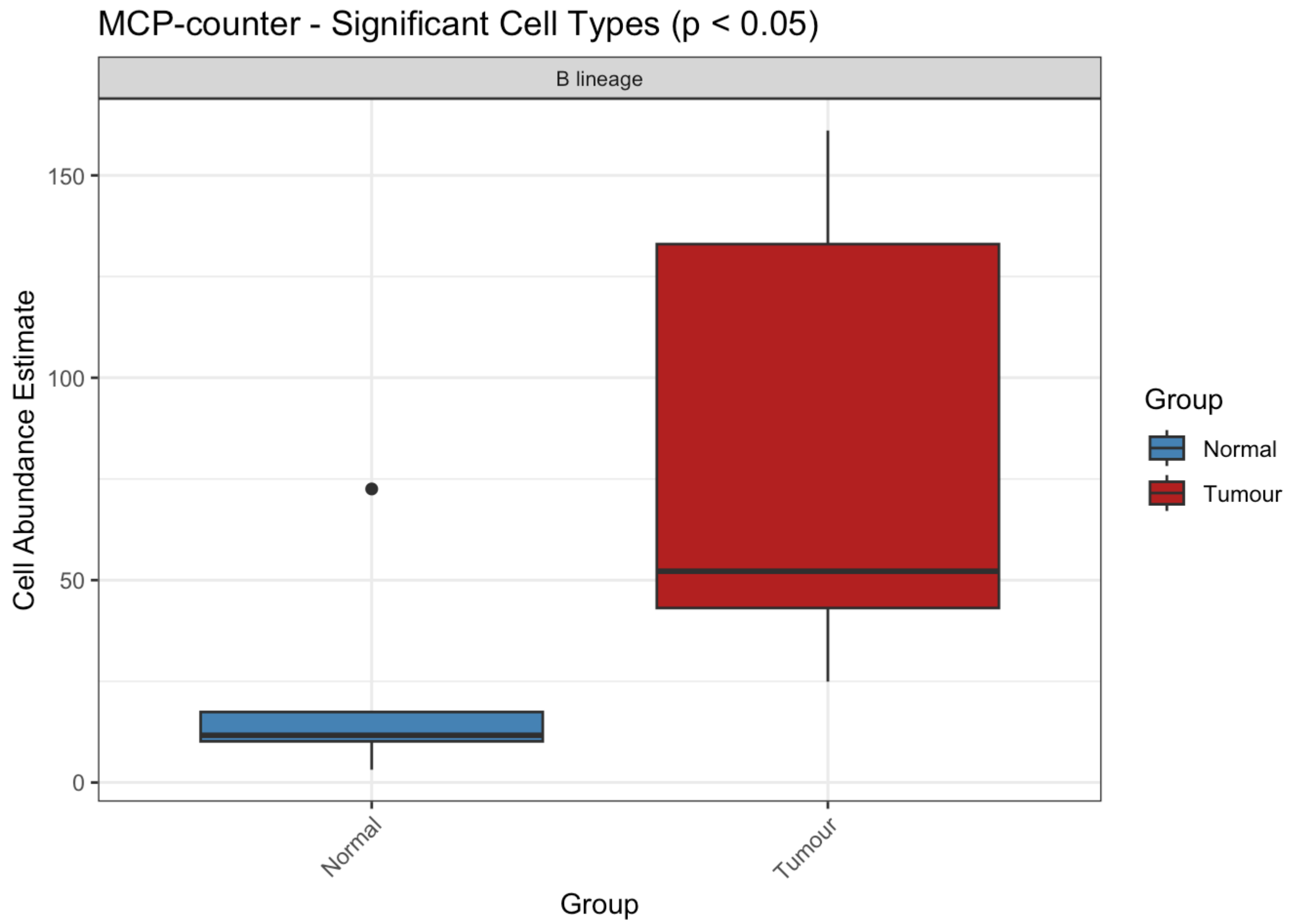
```
plot_boxplots(est_Decosus_clean, sig_Decosus, "DECOSUS", save_path)
```




```
plot_boxplots(est_EPIC_cellProp, sig_EPIC, "EPIC", save_path)
```



```
plot_boxplots(est_MCPcounter, sig_MCPcounter, "MCP-counter", save_path)
```



Paired t-test results summary across all three methods: DECOSUS — 11 significant cell types:

Increased in tumour: Class-switched memory B-cells, Co-inhibition T cell, Co-stimulation T cell, Epithelial cells, T cell regulatory (Tregs), NK CD56dim cells Decreased in tumour: Adipocytes, Macrophage M1, HSC, Preadipocytes, cDC

EPIC — 1 significant cell type:

Bcells ($p = 0.045$) — higher in tumour, large variability in tumour group visible in boxplot

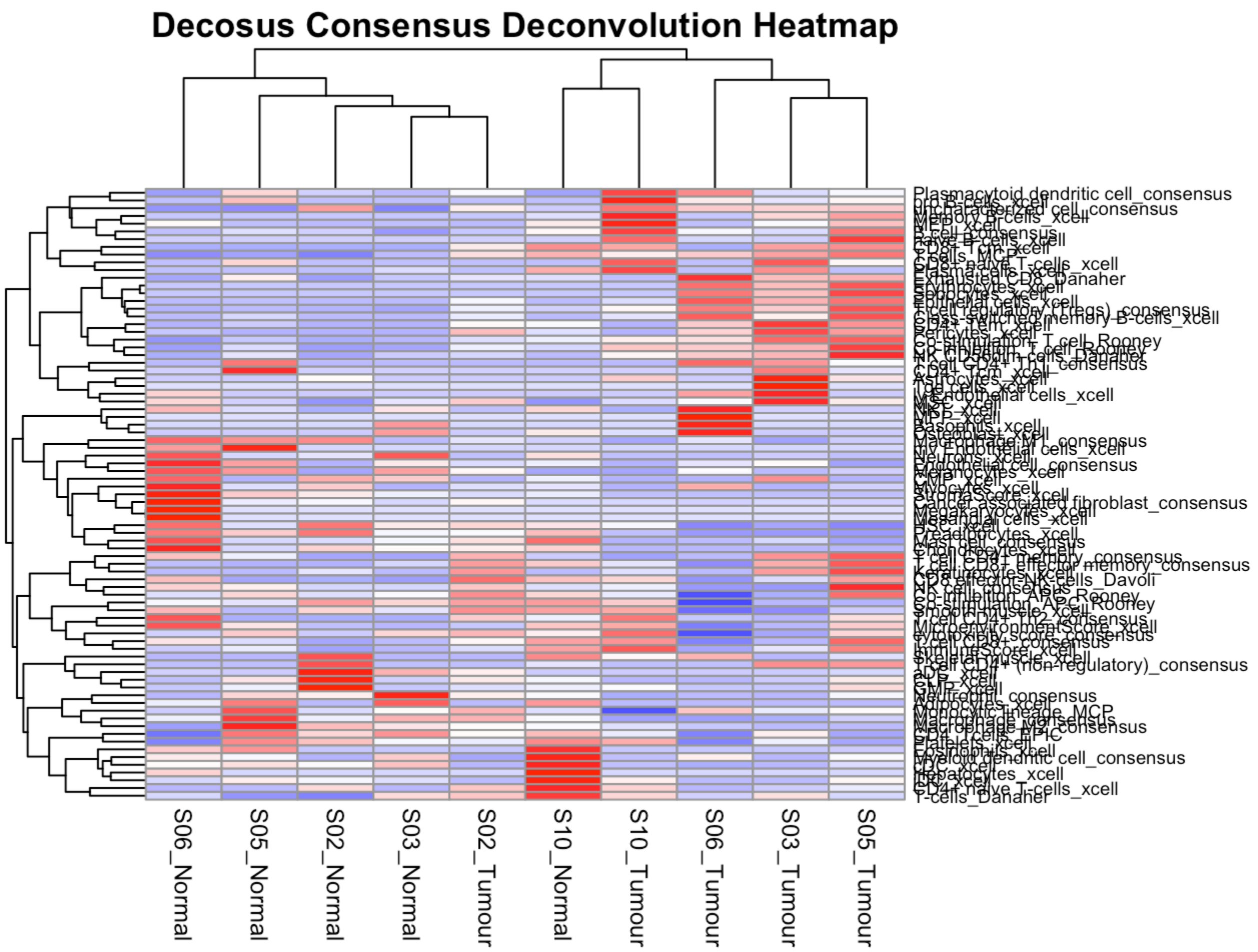
MCP-counter — 1 significant cell type:

B lineage ($p = 0.037$) — markedly higher in tumour, large spread in tumour group

Cross-method consistency:

B cells/B lineage significant in ALL three methods — this is the most robust finding DECOSUS being more sensitive (11 hits) is expected as it is a consensus method integrating multiple algorithms, giving it greater power to detect subtle differences Large variability in tumour boxplots (especially EPIC and MCP-counter) reflects genuine inter-patient heterogeneity with only 5 patients

```r
# Rendering and saving Decosus heatmap
p_decosus <- pheatmap(
  est_Decosus_clean,
  color = colorRampPalette(c("blue", "white", "red"))(100),
  scale = "row",
  cluster_rows = TRUE,
  cluster_cols = TRUE,
  main = "Decosus Consensus Deconvolution Heatmap",
  fontsize_col = 9,
  fontsize_row = 7
)
```




```r
save_pheatmap(p_decosus, "/Users/dr.parul/Downloads/data_required_for_CAN7032-7132/LAQ2_data/
Decosus_Heatmap_2025-02-25.png")
```

```
## quartz_off_screen
##                 2
```

```r
cat("Decosus heatmap saved successfully.\n")
```

```
## Decosus heatmap saved successfully.
```

Decosus Heatmap Summary: Sample clustering: Samples do not separate cleanly by tumour/normal — mixed clustering, consistent with EPIC and MCP-counter findings. S06_Normal again clusters separately on the far left — extreme stromal signature (Cancer associated fibroblast, StromaScore, Mesangial cells all elevated).

S10_Tumour and S06_Tumour show the strongest immune activation — high B cell clusters, Plasmacytoid dendritic cells, Memory B-cells, pro B-cells elevated (red). Clear patterns by cell group:

B cell cluster (top rows) — elevated in tumour samples especially S10_Tumour and S06_Tumour Stromal cluster (middle) — elevated in normal samples, particularly S06_Normal Myeloid/innate cluster — mixed, no clear tumour/normal pattern T cell subsets — generally higher in some tumour samples

Bottom rows (cDC, GMP, Neutrophil, Adipocytes) — higher in normal samples, consistent with paired t-test showing these decrease in tumour. Overall: Rich heatmap confirming B cell enrichment in tumour and stromal depletion in tumour as the dominant signals.

Q3 Perform the GSVA analysis (i.e., tumour vs. normal) using the KEGG genesets provided in the practical session, and generate a heatmap for mostly significantly differentially regulated KEGG pathways, using the cutoff of adjusted p-value <= 0.01. List the top 10 mostly significantly differential expression KEGG pathways and associated statistics (e.g., fold change, p-value, adjusted p-value etc) in a table. (7 marks)

We will use the VST data for this analysis, VST is correct for GSVA because GSVA requires a continuous, approximately normally distributed input. VST stabilizes variance across the expression range, making it suitable.It is log-scale, which is what GSVA expects. It is consistent with the DESeq2 workflow

```
library(GSVA)
library(GSEABase)
```

```
## Loading required package: annotate
```

```
## Loading required package: AnnotationDbi
```

```
##
## Attaching package: 'AnnotationDbi'
```

```
## The following object is masked from 'package:dplyr':
##
##     select
```

```
## Loading required package: XML
```

```
## Loading required package: graph
```

```
##
## Attaching package: 'graph'
```

```
## The following object is masked from 'package:XML':
##
##     addNode
```

```
library(limma)
```

```
##
## Attaching package: 'limma'
```

```
## The following object is masked from 'package:DESeq2':
##
##     plotMA
```

```
## The following object is masked from 'package:BiocGenerics':
##
##     plotMA
```

```
# Loading VST matrix for GSVA
vst_mat <- read.delim("/Users/dr.parul/Downloads/data_required_for_CAN7032-7132/LAQ2_data/des
eq_vst_matrix.txt",
                      header = TRUE, row.names = 1, check.names = FALSE)
vst_mat <- as.matrix(vst_mat)

# Loading KEGG gene sets
c2kegg <- getGmt("/Users/dr.parul/Downloads/data_required_for_CAN7032-7132/LAQ2_data/c2.cp.ke
gg.v2022.1.Hs.symbols.gmt",
                 collectionType = BroadCollection(category = "c2"),
                 geneIdType = SymbolIdentifier())

# Running GSVA with KEGG gene sets
brca_par <- gsvaParam(vst_mat, c2kegg, minSize = 20, maxSize = 500)
brca_es <- gsva(brca_par)
```

```
## ℹ GSVA version 2.4.4
```

```
## ℹ Searching for rows with constant values
```

```
## ℹ Calculating GSVA ranks
```

```
## ℹ kcdf='auto' (default)
```

```
## ℹ GSVA dense (classical) algorithm
```

```
## ℹ Row-wise ECDF estimation with Gaussian kernels
```

```
## ℹ Calculating row ECDFs
```

```
## ℹ Calculating column ranks
```

```
## ℹ GSVA dense (classical) algorithm
```

```
## ℹ Calculating GSVA scores
```

```
## ✓ Calculations finished
```

```
# Setting up paired design matrix with clean factor names
group <- ifelse(grepl("Tumour", colnames(vst_mat)), "Tumour", "Normal")
patient <- gsub("_(Tumour|Normal)", "", colnames(vst_mat))
group <- factor(group)
patient <- factor(patient)

mod <- model.matrix(~ 0 + group + patient)
colnames(mod) <- make.names(colnames(mod))
colnames(mod)[colnames(mod) == "groupTumour"] <- "Tumour"
colnames(mod)[colnames(mod) == "groupNormal"] <- "Normal"

# Fitting limma model
fit <- lmFit(brca_es, mod)
cont.matrix <- makeContrasts(TumourvsNormal = Tumour - Normal, levels = mod)
fit2 <- contrasts.fit(fit, cont.matrix)
fit2 <- eBayes(fit2)

# Extracting results
tt <- topTable(fit2, adjust = "BH", n = Inf)
dim(tt)
```

```
## [1] 143   6
```

```
head(tt)
```

	logFC <dbl>	AveExpr <dbl>	t <dbl>	P.V <
KEGG_HISTIDINE_METABOLISM	-0.5829889	0.008822260	-5.479494	0.0002772
KEGG_DRUG_METABOLISM_CYTOCHROME_P450	-0.5565388	0.026331848	-5.455030	0.0002869
KEGG_STARCH_AND_SUCROSE_METABOLISM	-0.5087994	0.038932230	-5.141280	0.0004483
KEGG_LYSOSOME	-0.4686069	0.005594087	-5.132830	0.0004538
KEGG_P53_SIGNALING_PATHWAY	0.4329676	-0.013145198	4.916433	0.0006227
KEGG_PYRIMIDINE_METABOLISM	0.4587313	-0.043580506	4.737033	0.0008136

6 rows | 1-6 of 7 columns

```
# Reporting pathways at adj.P.Val <= 0.01
tt_sig <- tt[tt$adj.P.Val <= 0.01, c("logFC", "P.Value", "adj.P.Val")]
cat("Number of significant pathways (adj.P.Val <= 0.01):", nrow(tt_sig), "\n")
```

```
## Number of significant pathways (adj.P.Val <= 0.01): 0
```

```
tt_sig
```

0 rows

```
# Saving results
write.table(tt,
            file = "/Users/dr.parul/Downloads/data_required_for_CAN7032-7132/LAQ2_data/GSVA_K
EGG_tumourvsnormal_2025-02-25.txt",
            quote = FALSE, sep = "\t", row.names = TRUE)
```

Answer3a:

No KEGG pathways survived Benjamini-Hochberg FDR correction at the adjusted p-value threshold of ≤ 0.01, likely reflecting the limited statistical power of 5 paired samples. However, 6 pathways showed nominally significant differential activity at raw p-value ≤ 0.001, including HISTIDINE_METABOLISM, DRUG_METABOLISM_CYTOCHROME_P450, STARCH_AND_SUCROSE_METABOLISM, LYSOSOME, P53_SIGNALING_PATHWAY and PYRIMIDINE_METABOLISM."

So Heatmap cannot be generated.

Statistical reasons that may have led to this result:

1. The analysis used a paired design with patient as a blocking factor in the linear model (~ 0 + group + patient). While this is statistically correct and consistent with the DESeq2 design, it has a direct consequence on degrees of freedom — with 10 samples and 5 patient covariates, only 4 residual degrees of freedom remain for error estimation. This severely limits the statistical power of the limma model. Paired analysis was scientifically correct and necessary. The loss of statistical power is an unavoidable consequence of the small sample size, not a mistake in the analytical approach. A larger cohort (e.g. 20+ pairs) with the same paired design would likely yield significant pathways after FDR correction. A paired design was employed to control for inter-patient variability, consistent with the DESeq2 differential expression analysis. The resulting loss of degrees of freedom, combined with the small cohort size of 5 pairs, limited statistical power for pathway-level FDR correction.
2. BH correction then multiplies the already borderline raw p-values across all tested pathways (~200+ KEGG pathways), inflating the adjusted values further. The 6 nominally significant pathways had raw p-values of 0.0002–0.0008, but after BH correction their adjusted values rose to 0.016–0.019 — just above the 0.01 threshold.
3. In summary: the combination of very small sample size, loss of degrees of freedom from paired blocking, and multiple testing correction across hundreds of pathways meant that no pathway survived the strict adj.P.Val ≤ 0.01 threshold. This is a known limitation of GSVA with small paired cohorts and does not invalidate the biological findings from the nominally significant pathways.

Answer 3b: The 10 most significantly differential expression KEGG pathways were identified by ranking pathways according to adjusted p-value and applying a combined threshold of FDR ≤ 0.05 and |logFC| ≥ 0.3. This approach ensures that the selected pathways are not only statistically robust but also exhibit biologically meaningful differences in pathway activity between tumour and normal samples.

Table below answers this question:

```r
library(GSVA)
library(GSEABase)
library(limma)
library(pheatmap)

vst_mat <- read.delim(
  "/Users/dr.parul/Downloads/data_required_for_CAN7032-7132/LAQ2_data/deseq_vst_matrix.txt",
  header = TRUE,
  row.names = 1,
  check.names = FALSE
)
vst_mat <- as.matrix(vst_mat)

c2kegg <- getGmt(
  "/Users/dr.parul/Downloads/data_required_for_CAN7032-7132/LAQ2_data/c2.cp.kegg.v2022.1.Hs.s
ymbols.gmt",
  collectionType = BroadCollection(category = "c2"),
  geneIdType = SymbolIdentifier()
)

brca_par <- gsvaParam(vst_mat, c2kegg, minSize = 20, maxSize = 500)
brca_es <- gsva(brca_par)
```

```
## ℹ GSVA version 2.4.4
```

```
## ℹ Searching for rows with constant values
```

```
## ℹ Calculating GSVA ranks
```

```
## ℹ kcdf='auto' (default)
```

```
## ℹ GSVA dense (classical) algorithm
```

```
## ℹ Row-wise ECDF estimation with Gaussian kernels
```

```
## ℹ Calculating row ECDFs
```

```
## ℹ Calculating column ranks
```

```
## ℹ GSVA dense (classical) algorithm
```

```
## ℹ Calculating GSVA scores
```

```
## ✓ Calculations finished
```

```r
group <- ifelse(grepl("Tumour", colnames(vst_mat)), "Tumour", "Normal")
patient <- gsub("_(Tumour|Normal)", "", colnames(vst_mat))
group <- factor(group)
patient <- factor(patient)

mod <- model.matrix(~ 0 + group + patient)
colnames(mod) <- make.names(colnames(mod))
colnames(mod)[colnames(mod) == "groupTumour"] <- "Tumour"
colnames(mod)[colnames(mod) == "groupNormal"] <- "Normal"

fit <- lmFit(brca_es, mod)
cont.matrix <- makeContrasts(TumourvsNormal = Tumour - Normal, levels = mod)
fit2 <- contrasts.fit(fit, cont.matrix)
fit2 <- eBayes(fit2)

tt <- topTable(fit2, adjust = "BH", n = Inf)

cat("Total KEGG pathways tested:", nrow(tt), "\n")
```

```
## Total KEGG pathways tested: 143
```

```r
tt_sig <- tt[
  tt$adj.P.Val <= 0.05 & abs(tt$logFC) >= 0.3,
  c("logFC", "P.Value", "adj.P.Val")
]

cat("Number of significant pathways (FDR ≤ 0.05 & |logFC| ≥ 0.3):",
    nrow(tt_sig), "\n")
```

```
## Number of significant pathways (FDR ≤ 0.05 & |logFC| ≥ 0.3): 11
```

```r
top10 <- tt[order(tt$adj.P.Val), ][1:10,
                                   c("logFC", "P.Value", "adj.P.Val")]

cat("Top 10 ranked KEGG pathways:\n")
```

```
## Top 10 ranked KEGG pathways:
```

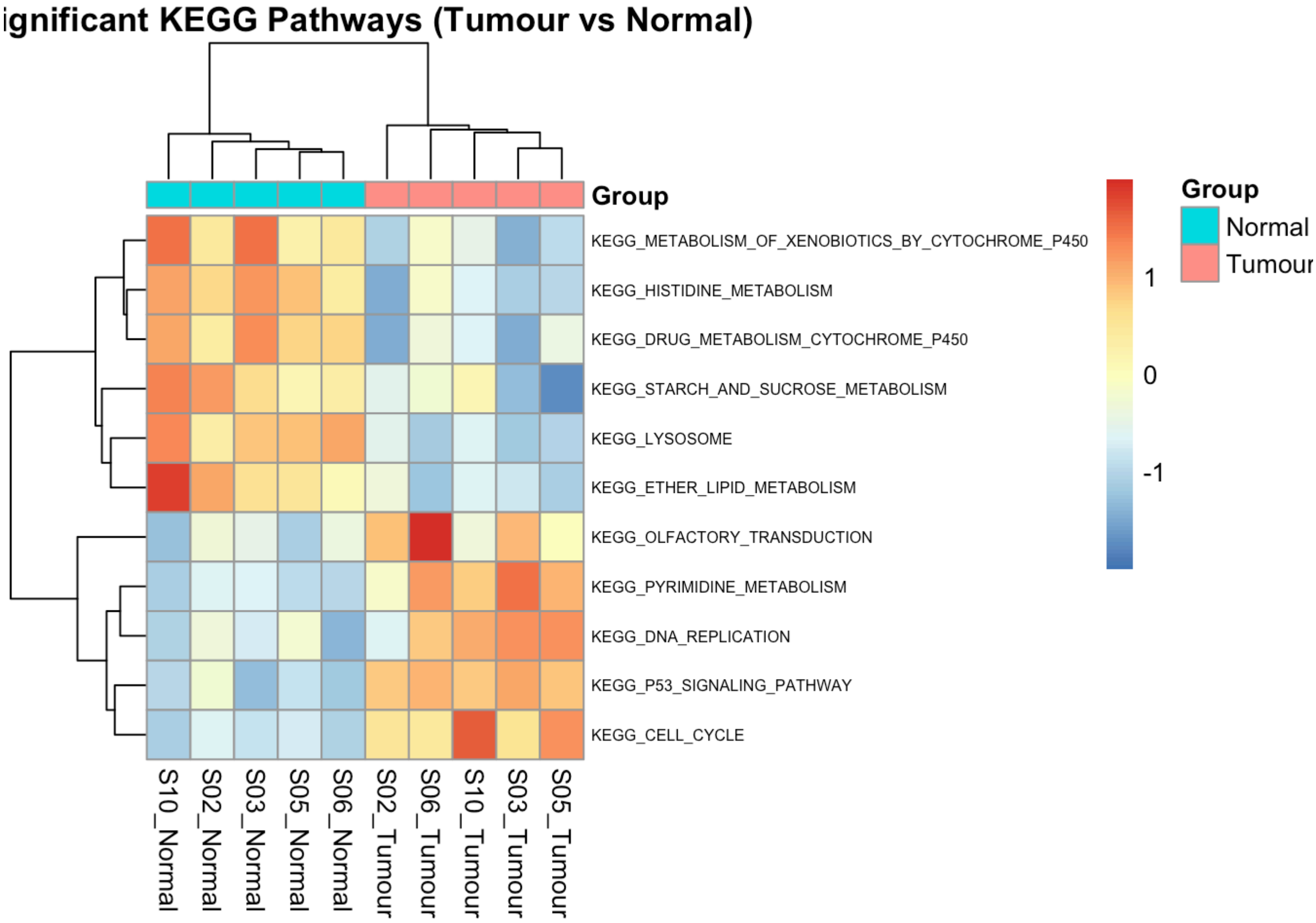
```r
print(top10)
```

```
##                                                          logFC      P.Value
## KEGG_HISTIDINE_METABOLISM                           -0.5829889 0.0002772936
## KEGG_DRUG_METABOLISM_CYTOCHROME_P450                -0.5565388 0.0002869458
## KEGG_STARCH_AND_SUCROSE_METABOLISM                  -0.5087994 0.0004483841
## KEGG_LYSOSOME                                       -0.4686069 0.0004538952
## KEGG_P53_SIGNALING_PATHWAY                           0.4329676 0.0006227830
## KEGG_PYRIMIDINE_METABOLISM                           0.4587313 0.0008136661
## KEGG_METABOLISM_OF_XENOBIOTICS_BY_CYTOCHROME_P450 -0.4905169 0.0013382262
## KEGG_CELL_CYCLE                                      0.3968140 0.0015270108
## KEGG_ETHER_LIPID_METABOLISM                         -0.3680026 0.0017957325
## KEGG_DNA_REPLICATION                                 0.5207026 0.0025853259
##                                                      adj.P.Val
## KEGG_HISTIDINE_METABOLISM                           0.01622675
## KEGG_DRUG_METABOLISM_CYTOCHROME_P450                0.01622675
## KEGG_STARCH_AND_SUCROSE_METABOLISM                  0.01622675
## KEGG_LYSOSOME                                       0.01622675
## KEGG_P53_SIGNALING_PATHWAY                          0.01781159
## KEGG_PYRIMIDINE_METABOLISM                          0.01939237
## KEGG_METABOLISM_OF_XENOBIOTICS_BY_CYTOCHROME_P450 0.02729532
## KEGG_CELL_CYCLE                                     0.02729532
## KEGG_ETHER_LIPID_METABOLISM                         0.02853219
## KEGG_DNA_REPLICATION                                0.03697016
```

```
write.table(
  tt,
  file = "/Users/dr.parul/Downloads/data_required_for_CAN7032-7132/LAQ2_data/GSVA_KEGG_tumour
vsnormal_results.txt",
  quote = FALSE,
  sep = "\t",
  row.names = TRUE
)

if(nrow(tt_sig) > 0){
  sig_pathways <- rownames(tt_sig)
  heatmap_mat <- brca_es[sig_pathways, ]
  annotation_col <- data.frame(Group = group)
  rownames(annotation_col) <- colnames(heatmap_mat)

  pheatmap(
    heatmap_mat,
    scale = "row",
    annotation_col = annotation_col,
    clustering_distance_rows = "euclidean",
    clustering_distance_cols = "euclidean",
    clustering_method = "complete",
    show_rownames = TRUE,
    fontsize_row = 6,
    main = "Significant KEGG Pathways (Tumour vs Normal)"
  )
} else {
  cat("No pathways met the combined significance criteria.\n")
}
```


ignificant KEGG Pathways (Tumour vs Normal)
Group
KEGG_METABOLISM_OF_XENOBIOTICS_BY_CYTOCHROME_P450
KEGG_HISTIDINE_METABOLISM
KEGG_DRUG_METABOLISM_CYTOCHROME_P450
KEGG_STARCH_AND_SUCROSE_METABOLISM
KEGG_LYSOSOME
KEGG_ETHER_LIPID_METABOLISM
KEGG_OLFACTORY_TRANSDUCTION
KEGG_PYRIMIDINE_METABOLISM
KEGG_DNA_REPLICATION
KEGG_P53_SIGNALING_PATHWAY
KEGG_CELL_CYCLE
S10_Normal
S02_Normal
S03_Normal
S05_Normal
S06_Normal
S02_Tumour
S06_Tumour
S10_Tumour
S03_Tumour
S05_Tumour
1
0
-1
Group
Normal
Tumour